Personalized Food Assistant: A Location-Based Prescriptive Chatbot for Food Recommendations

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1. INTRODUCTION

1.1 Project Context

Choosing what to eat and where to buy food can be a time consuming process, especially with so many options. Consumers are using their smartphones to find restaurants, book tables and read reviews so there's a need for more intelligent recommendation services. Businesses and public institutions are adding AI powered chatbots to their platforms to improve service and user engagement. For example AI chatbots have been shown to improve public services by providing 24/7 access to information and simplifying interactions and increasing user satisfaction [1]. AI allows food recommender systems to analyze user behavior, identify consumption patterns and provide personalized meal suggestions that can adapt over time [2].

Current food recommender systems provide restaurant listings, customer reviews and food delivery services to help users explore dining options around them. Some applications also offer rating systems and filters for price, cuisine type, and user preferences. However, these systems often lack personalization, as they do not fully consider individual user preferences, dietary restrictions, and health conditions. This results in meal recommendations that may not always align with users' specific needs, leading to frustration and inefficient decision-making when choosing meals. A recent approach has utilized structured knowledge through an ontology-based chatbot to deliver healthy food suggestions specifically for teenagers, enhancing dietary decisions through more targeted recommendations [3].

In a survey conducted with 50 respondents, survey results indicate that 72% of respondents struggle to find meal recommendations that match their dietary restrictions and health conditions, 64% of users rely on online food recommendations but often find them inaccurate, 59% prefer food suggestions based on currently open establishments near them, and 67% believe that AI-driven recommendations could improve decision-making efficiency. These findings reinforce the gap in existing food recommendation systems, highlighting the need for a solution that offers personalized, adaptive, and context-aware meal suggestions. Additionally, existing chatbot APIs offer basic conversational features but cannot customize interactions according to specific dietary requirements and health issues. Health-aware food recommendation systems based on knowledge graph reasoning have demonstrated the importance of considering dietary restrictions and health conditions in meal suggestions [4]. A chatbot system that refines its suggestions based on real-time user feedback has shown to improve user satisfaction and engagement in restaurants [5].

Table 1. Features Comparison Matrix

Features	FoodPanda	GrabFood	PLBFA
Personalized Meal Recommendations	✗	✗	✓
Location-Based Suggestions	✓	✓	✓
Health-Conscious Options	Limited	Limited	✓
User Preference Learning	✗	✗	✓
Restaurant Ratings & Reviews	✓	✓	✓
Delivery Services	✓	✓	✗
Dietary Filters (e.g., vegan)	Limited	Limited	✓
Platform Accessibility	Mobile App and Web	Mobile App and Web	Web
Focus on Local Businesses	✓	✓	✓
Integration with Maps/Geolocation	✓	✓	✓
Ease of Use	Moderate	Moderate	High
Cost to Users	Delivery Fees Apply	Delivery Fees Apply	Free to Use
Primary Users	Delivery-focused customers	Delivery-focused customers	Time-conscious, health-conscious, and fitness-conscious individuals
Google AdSense Monetization	✗	✗	✓
Sponsored Content & Promotions	✓	✓	✓

Table 1 compares existing food delivery platforms (FoodPanda, GrabFood) and the proposed Personalized Food Assistant (PFA). The table highlights the unique features of PLBFA - personalized meal recommendations, health conscious options, user preference learning and prescriptive analytics that differentiates it from traditional delivery focused platforms.

By integrating AI and Google Dialogflow based NLP, the Personalized Location-Based Food Assistant aims to improve existing food recommendation systems by considering dietary restrictions, cultural preferences and real-time contextual information. This integration will provide users with more relevant meal suggestions and address the limitations of traditional food recommender systems. If successful, the system will help users make better dining decisions and explore more options more efficiently [6].

With geolocation technology the system can provide meal recommendations based on where you are. By accessing real time position data the chatbot can suggest places to eat near you within a customizable radius so you can find options quickly. To make sure the location based recommendations are accurate and relevant the system uses the Haversine formula, a mathematical algorithm that calculates the distance between two points on the surface of the Earth based on their latitude and longitude coordinates. This is the same formula used in many geolocation services including the Google Maps API which powers location based search and routing. By putting this formula into the backend logic of the chatbot it can pinpoint and recommend food places that are actually near the user. This approach addresses the challenge of users spending excessive time searching for suitable meal venues by offering accurate, location-specific suggestions. Location-based services have been shown to enhance user engagement by providing contextually relevant recommendations [7].

This study was done in Davao City, Philippines targeting young professionals, students and time-conscious individuals who use food delivery or meal planning apps. They are the primary users and beneficiaries of this AI-powered food recommendation system. The platform will aid these users by providing them with smarter, faster, and healthier meal recommendations that align with their preference and location. Making healthier food choices through personalized AI-driven meal recommendations: a qualitative study of health-seeking behavior Not Applicable This study aligns with UN Sustainable Development Goal (SDG) 3: Good Health and Well-being, as it facilitates users (image a patient or caregiver seeking a healthy meal by being able to provide a guide that suggests healthier meal options that are in line with the user's Dietary Requirements and Preferences.

Unlike other food recommender systems that use predefined filters, this study uses AI powered natural language processing (NLP) models with Google Dialogflow to interpret user queries and adapt recommendations in real time. Instead of constraining recommended meals to static keywords, it allows natural language input and extracts implicit dietary preferences, health issues, and behavior patterns from interacting with the system. Through users' feedback and past interactions, the chatbot iterates on its knowledge and formulates personalized suggestions that adjust to new preferences and eating habits. By leveraging this adaptive NLP system the meal suggestions can take on a truly proactive, as opposed to reactive approach to serve and support individual needs, which standard static filtering mechanisms in most food recommendation systems cannot provide. To achieve this, the study will use Google Dialogflow API as the conversational engine. The chatbot's NLP capabilities via Google Dialogflow will be managed through C# and ASP.NET Core backend integration to restrict responses to only food related queries and filter out unrelated prompts. This way the chatbot will remain contextually relevant and only provide meal related recommendations, making it more usable and effective for the user. NLP in AI driven food recommendation systems has been shown to improve personalization by analyzing user provided data through Dialogflow and ASP.NET Core [8]. An AI chatbot based system using personalization data has also been proven to tailor meal suggestions to individual behavior and preferences [9].

To provide actionable and optimal meal suggestions, this study uses a prescriptive recommendation approach. Unlike descriptive or

predictive systems, a prescriptive recommender system not only looks at user preferences and context but also proactively suggests the best meal options to help users achieve their dietary and health goals. It does this through behavioral learning - the product records and interprets what the user does in terms of what they eat, how they rate recommendations, recurrence of specific types of food and even time-based habits. These data points are processed using AI to find emerging patterns and to refine and personalize recommendations over time. Prescriptive recommendation systems have been shown to work well in health and nutrition applications [10].

The use of Artificial Intelligence (AI) in this system will overcome the limitations of existing food recommender platforms. AI will process user behaviour, dietary preferences and feedback to find consumption patterns. Over time this will help the chatbot to refine its recommendations making them more relevant to the user's changing tastes and health needs. Unlike traditional suggestion systems, AI will provide dynamic recommendations by taking into account individual user characteristics and contextual factors like location [11].

The chatbot will be web based and users will be able to get meal recommendations online. To support the system's sustainability, Google AdSense and sponsored content will be considered. AI powered apps can also provide real time dietary feedback and health insights to help users make informed food choices [12].

1.2 Purpose and Description

The purpose of this project is to introduce an intelligent web based food assistant that uses prescriptive recommendation methods to solve the problem of meal selection and food buying. By using artificial intelligence, natural language processing and geolocation, the system goes beyond personalization by proactively guiding users to meal options and nearby places that fit their dietary needs, preferences and situational context. Through behavioral learning the assistant will optimize user decision making, to help users make healthier and more efficient food choices. Globally this research contributes to the growing body of research in AI driven nutrition systems and prescriptive analytics, to the digital health technologies. Socially, it aligns with the United Nations Sustainable Development Goal 3: Good Health and Well-being and supports the CCE research agenda on technology-driven community solutions by promoting accessible, health-conscious decision-making for time-conscious individuals, students, and working professionals.

This project, titled "Personalized Location-Based Food Assistant: A Prescriptive Recommender Chatbot for Meal Recommendation and Food Purchase," integrates Google Dialogflow-based natural language processing and geolocation services within a web-based platform built on ASP.NET Core and C#. The system analyzes user inputs, dietary preferences, health conditions, and real-time location to generate actionable meal suggestions and recommend nearby food establishments. To make location based recommendations accurate and relevant the system uses the Haversine formula. This math formula can be used to determine the shortest distance between two points that are the surface of the earth from latitude and longitude. This allows the chatbot to pinpoint and recommend food places within a user defined radius and make the suggestions more contextual. By using behavioral learning the chatbot refines its recommendations so suggestions are not only personalized but also prescriptive to the user's evolving needs and context. This addresses the limitations of traditional food recommendation systems and brings innovative solutions to the fields of AI, nutrition and digital health.

1.3 Objectives

1.3.1 General Objectives

This study aims to build and integrate an AI chatbot using Google Dialogflow for NLP and geolocation services in a web based platform. The system will help users to make better decisions by providing personalized meal recommendations based on individual dietary preferences, health requirements and nearby food places, proximity calculated using Haversine formula.

1.3.2 Specific Objectives

To achieve the general objectives, the application aims to:

1.3.2.1 Integrate an AI chatbot using Google Dialogflow API to process natural language user inputs through C# and ASP.NET Core backend to understand dietary restrictions, health conditions and preferences for generating personalized meal recommendations.

1.3.2.2 Integrate Google Maps API geolocation services using JavaScript and ASP.NET Core to detect users' real-time location coordinates, apply the Haversine formula to calculate distances to nearby food establishments, and filter recommendations based on customizable radius parameters.

1.3.2.3 Implement a dietary and health restriction filtering system using SQL Server database tables to store user dietary preferences (vegetarian, vegan, gluten-free, etc.), allergies, and health conditions, enabling the chatbot to filter meal recommendations accordingly.

1.3.2.4 Modify the chatbot's recommendation system by implementing user preference tracking using Entity Framework Core to store and analyze user interaction data, enabling the system to improve recommendation accuracy through pattern recognition, prescriptive recommendation, and preference learning algorithms.

1.4 Scope and Limitations

This study encompasses the development and deployment of a web-based, AI-powered chatbot that provides prescriptive, personalized meal recommendations for users in Davao City, Philippines. The system utilizes Google Dialogflow for natural language processing, Google Maps API for geolocation, and behavioral learning algorithms to suggest meals and nearby food establishments based on individual dietary preferences, health considerations, and real-time location. The recommendation engine employs a five-parameter weighted scoring model with empirically-derived weight allocations based on established research in health-aware recommendation systems. The parameter weighting follows evidence-based principles with strong academic justification: (1) Dietary Restriction Weight (25%) receives highest priority as safety constraints are non-negotiable in health-aware

systems, with research emphasizing the importance of filtering meals based on health conditions and dietary restrictions to ensure user safety [13]; (2) User Preference Weight (30%) maintains high 0 importance as explicit preferences are crucial for user satisfaction and system adoption, with studies demonstrating that multi-objective frameworks effectively balance user preferences with health constraints in personalized food recommendations [14]; (3) Nutritional Optimization Weight (20%) supports long-term health outcomes, with research indicating the effectiveness of multi-objective optimization approaches that jointly optimize user preference, personalized healthiness, and nutritional diversity [15]; (4) Health Condition Weight (10%) provides medical consideration while recognizing the importance of personalized meal planning for individuals with diet-related health concerns [16]; and (5) Behavioral Pattern Weight (15%) enables personalization through adaptive learning from user interactions, with studies showing the effectiveness of systems that capture dynamic user behaviors and learn individual food preferences from interaction patterns [17]. The system's location-based recommendation engine leverages the Haversine formula to accurately compute the distance between users and food establishments, ensuring that only those within a specified radius are suggested. The project includes the integration of user input analysis, adaptive prescriptive recommendation algorithms, and a responsive user interface built with ASP.NET Core and Tailwind CSS. The study is conducted within the academic year 2024–2025 and targets young professionals, students, and time-conscious individuals, specifically within the age range of 18 to 40 years old, who frequently use digital platforms for meal planning and food discovery.

The project is limited by several factors beyond the researcher's control. The accuracy of meal recommendations depends on the quality and completeness of user-provided information and the availability of up-to-date data on local food establishments. The system's performance may be affected by internet connectivity, third-party API reliability (such as Google Dialogflow and geolocation services), and the scope of the dataset used for training and recommendations. Additionally, while the HAVERSINE FORMULA provides an effective method for calculating straight-line distances, it does not account for real-world travel routes or obstacles, which may affect the actual accessibility of recommended establishments. The chatbot does not provide medical or nutritional advice beyond general recommendations and may not fully address complex dietary or health conditions. The study does not cover large-scale deployment or integration with food delivery services, but focuses on developing responsive web interfaces for both desktop and mobile platforms.

2. METHODOLOGY

The development of the "Personalized Location-Based Food Assistant: A Prescriptive Recommender Chatbot for Meal Recommendation and Food Purchase" followed the Agile methodology, specifically the Scrum framework, which enabled the project team to iteratively design, develop, and refine the system through structured sprints. The team consisted of a Product Owner (Team Leader), Scrum Master, and a Development Team composed of AI engineers, software developers, and system testers, with key Scrum ceremonies such as Sprint Planning, Daily Standups, Sprint Reviews, and Sprint Retrospectives conducted to ensure continuous improvement and alignment with user needs.

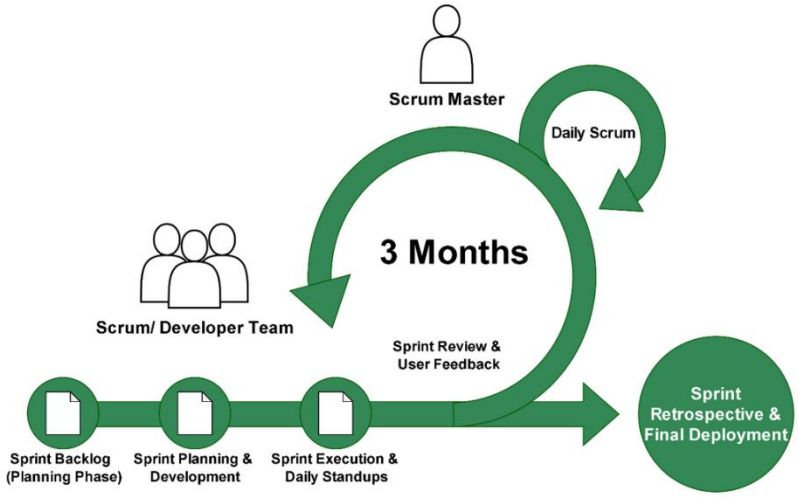


Figure 1 Agile Scrum

The technical implementation utilized ASP.NET Core and C# for backend development, Tailwind CSS for responsive user interface design, and Google Dialogflow for natural language processing, while the system's database was managed using Microsoft SQL Server with Entity Framework Core handling object-relational mapping. Geolocation features were integrated using the Google Maps API, and the Haversine formula was implemented to calculate distances between users and food establishments, enabling the chatbot to provide location-based meal recommendations. The prescriptive recommendation engine leveraged behavioral learning algorithms to continuously refine and optimize meal suggestions based on user interactions and preferences.

The project was structured into several distinct phases: the **Project Initiation and Requirement Gathering** phase involved identifying project objectives, defining the system scope, and gathering requirements from stakeholders and potential users through surveys and interviews to understand user needs regarding meal recommendations, dietary restrictions, and location-based

services, with the product backlog created to outline all required features and tasks.

The **Chatbot and NLP API Development** phase focused on integrating Google Dialogflow with the ASP.NET Core backend to enable natural language processing, designing conversational flows, intents, and entities to ensure the chatbot could interpret user queries related to food preferences, health conditions, and location, with initial testing conducted to validate the chatbot's ability to process and respond to user inputs accurately.

The **AI and Personalization Integration** phase involved developing the recommendation logic by implementing algorithms to analyze user profiles, preferences, and behavioral data, using Entity Framework Core to manage user data in the SQL Server database, and configuring the prescriptive recommendation engine to generate meal suggestions tailored to individual users.

The **Geolocation Services Implementation** phase focused on developing the geolocation module by integrating the Google Maps API and implementing the Haversine formula for distance calculation, configuring the system to detect user location, retrieve nearby food establishments, and filter recommendations based on proximity, with testing ensuring that location-based suggestions were accurate and relevant.

The **Prescriptive Recommendation System Optimization** phase involved refining the recommendation engine by incorporating behavioral learning algorithms, analyzing user interactions and feedback to improve the accuracy and relevance of meal suggestions, and adjusting the system to adapt to changing user preferences over time.

The User Feedback Collection phase had mechanisms for collecting user feedback including rating systems and feedback forms, monitoring user responses to recommendations to identify areas for improvement and using feedback to update the product backlog and inform future development cycles.

Finally, the Final Deployment phase involved deploying the system to a production environment, conducting final testing and addressing any remaining issues, completing documentation, making the system available to users, and establishing post-deployment monitoring to track system performance and user satisfaction.

2.1 Data Gathering

Data for this project were collected using a structured Google Form survey, which was distributed through social media platforms such as Facebook Messenger. The survey was designed to gather both quantitative data (such as frequency of meal planning, use of food apps, and demographic information) and qualitative insights (such as personal challenges, preferences, and suggestions for improvement) from respondents. Target participants included young professionals, students, and other individuals in Davao City who regularly use digital platforms for meal planning and food discovery, representing the primary beneficiaries of the proposed system.

Participants were invited to complete the survey via direct messages and group chats, ensuring a diverse and relevant sample. The Google Form included multiple-choice, Likert scale, and open-ended questions to capture a comprehensive understanding of user needs and expectations. All responses were automatically recorded and organized within Google Sheets for subsequent analysis, which informed the definition of user requirements and guided the prioritization of system features.

Ethical considerations were strictly observed throughout the data gathering process. Participation was voluntary, and respondents were informed about the purpose of the study and the confidentiality of their responses. No personally identifiable information was collected without consent, and all data was handled in compliance with privacy and ethical standards to ensure the protection of participants' rights and information.

2.2 Analysis

2.2.1 User Requirements

To ensure the successful development and adoption of the Personalized Location-Based Food Assistant, it is essential to identify and document the specific needs of the target users. These requirements are based on the roles and expectations of the system's primary beneficiaries, including:

2.2.1.1. End Users (Young Professionals, Students, Time-Conscious Individuals)

End users require a user-friendly interface accessible via both desktop and mobile devices for convenient meal planning, and they need personalized meal recommendations that consider dietary restrictions, health conditions, and food preferences. In addition, they desire real-time suggestions for nearby food establishments based on their current location, and they expect quick and relevant responses from the chatbot, with the ability to provide feedback on recommendations. The system's ability to recommend establishments within a specific radius is powered by the Haversine formula, ensuring that only truly nearby options are presented to the user.

2.2.1.2 System Administrators

System administrators need tools to manage and update the database of food establishments, menu items, and user profiles, and they require access to analytics and user feedback to monitor system performance and improve service quality. Moreover, they must ensure data privacy and security for all user information.

2.2.1.3 User Preferences

Based on survey responses and interviews, users requested additional features, including the ability to save preferred restaurants and meals, filter recommendations by price range and cuisine type, receive notifications about new restaurants in their area, and provide ratings and feedback on recommendations through the chatbot interface. Users also expressed interest in customizing their dietary preferences with specific allergies, intolerances, and health conditions, and the option to view basic nutritional information for recommended meals when available.

2.2.2 Hardware and Software Requirements

2.2.2.1 Hardware Requirements

2.2.2.1.1 Web Servers

Reliable server infrastructure to host the web application and database, ensuring high availability and performance. The server must support concurrent user connections and ensure 99.5% uptime for reliable system availability. Cloud-based hosting solutions or dedicated servers are recommended for production deployment.

2.2.2.1.2 User Devices

Desktop computers and laptops with Windows 10/11, macOS, or Linux operating systems, equipped with a minimum of 4GB RAM and modern web browsers (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+). Mobile devices, including smartphones (iOS 14+, Android 8+) and tablets with responsive web browser support for cross-platform accessibility.

2.2.2.2 Software Requirements

2.2.2.2.1 Backend

ASP.NET Core 8.0 framework with C# programming language for developing RESTful APIs, middleware components, and server-side business logic. Includes Entity Framework Core for database operations, dependency injection for service management, and built-in security features for authentication and authorization.

2.2.2.2.2 Frontend

Tailwind CSS 3.0+ for utility-first CSS framework implementation, enabling responsive design across multiple screen sizes. Includes custom CSS components, JavaScript for interactive elements, and HTML5 semantic markup for accessible user interface development.

2.2.3 Functional and Non-Functional Requirements

2.2.3.1. Functional Requirements

2.2.3.1.1 User Registration and Authentication

Secure user registration system with email verification, password hashing using bcrypt, and JWT token-based authentication. Includes user profile management, session handling, and role-based access control for administrators and regular users.

2.2.3.1.2 Personalized Meal Recommendations

AI-powered chatbot utilizing Google Dialogflow for natural language understanding, processing user queries to extract dietary preferences, health conditions, and meal requirements. Generates prescriptive recommendations based on user profile data, historical interactions, behavioral patterns, and available restaurant information, ensuring optimal meal suggestions that align with user goals and preferences.

2.2.3.1.3 Geo-location Based Search

Real-time location detection using browser geolocation API, integration with Google Maps API for nearby restaurant identification within a configurable radius (default 2-5km). The system applies the Haversine formula to calculate the straight-line distance between the user's current coordinates and each establishment, filtering recommendations to include only those within the specified radius. This ensures that users receive contextually relevant and accessible meal options.

2.2.3.1.4 User Feedback Collection

Interactive rating system (1-5 stars) for restaurant and meal recommendations, text-based feedback forms, and preference adjustment mechanisms. Stores feedback data for recommendation algorithm improvement and user experience enhancement.

2.2.3.1.5 Chatbot Experience and Restaurant Discovery Prompt

The chatbot will automatically prompt users to share feedback about their experience after receiving meal recommendations. It will also ask whether the user has successfully found a nearby restaurant to suggest, enabling the system to gather insights for improving location-based accuracy and user satisfaction.

2.2.3.2 Non-Functional Requirements

2.2.3.2.1 Usability

Intuitive user interface with clear navigation, consistent design patterns, and accessibility features (WCAG 2.1 compliance). Supports users with varying technical expertise through guided interactions and contextual help features.

2.2.3.2.2 Scalability

Horizontal scaling capability through load balancing, database optimization with indexing strategies, and caching mechanisms (Redis/Memory Cache). Designed to handle concurrent user sessions and increasing data volume without performance degradation.

2.2.3.2.3 Security

Data encryption for sensitive information, HTTPS protocol enforcement, SQL injection prevention through parameterized queries, and cross-site scripting (XSS) protection. Implements secure authentication, authorization, and data privacy compliance measures.

2.2.3.2.4 Performance

Provides fast response times and minimal downtime for an ideal user experience. Optimized database queries, efficient caching strategies, and CDN integration for static content delivery to ensure a responsive user experience.

2.2.3.2.5 Cross-Platform Compatibility

Responsive web design using Tailwind CSS breakpoints (sm, md, lg, xl) for functionality across desktop browsers (Chrome, Firefox, Safari, Edge) and mobile devices (iOS Safari, Android Chrome). Progressive Web App (PWA) features for enhanced mobile experience.

2.2.4 Feasibility Study

2.2.4.1 Technical Feasibility

The chosen technologies—ASP.NET Core, C#, Microsoft SQL Server, Tailwind CSS, Google Dialogflow, and Google Maps

API—are well-established and widely supported, ensuring a robust and scalable foundation for the system. The integration of the HAVERSINE FORMULA for geospatial distance calculation further strengthens the technical feasibility by providing a reliable method for implementing location-based filtering. The development team possesses the necessary expertise to implement and integrate these technologies effectively.

2.2.4.2 Operational Feasibility

The system addresses the critical needs of the target users by providing prescriptive, personalized, location-based meal recommendations and a user-friendly interface. Agile Scrum ensures continuous improvement and alignment with user feedback.

2.2.4.3 Economic Feasibility

Initial development and integration costs yes, but long term benefits (user satisfaction, healthier eating, potential for future monetization) justify the investment. The system is scalable and adaptable which makes it economically viable.

2.2.4.4 Schedule Feasibility

Structured Agile sprints and milestones will make the project a done deal within the semester. The incremental approach to development enables incremental delivery, risk management and refinement, responding to feedback on behalf of the user and stakeholder.

Table 2 Project Duration Timeline

Activities	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
Data Gathering										
Title Proposal										
Requirements Analysis										
Outline Defense										
Systems Development										
Systems Testing										
Final Defense										

The project timeline spans ten months from January to October, with activities strategically distributed across different phases of development. The initial phase focuses on foundational work, including data gathering and title proposal development from January to March, followed by requirements analysis extending through May. The outline defense is scheduled for June, marking the transition to the development phase. Systems development represents the most intensive period, running from July through September, while systems testing overlaps in the final month of development to ensure quality assurance. The project concludes with the final defense scheduled for October, allowing adequate time for system refinement and documentation completion before presentation.

2.3 Design

2.3.1 Conceptual Framework

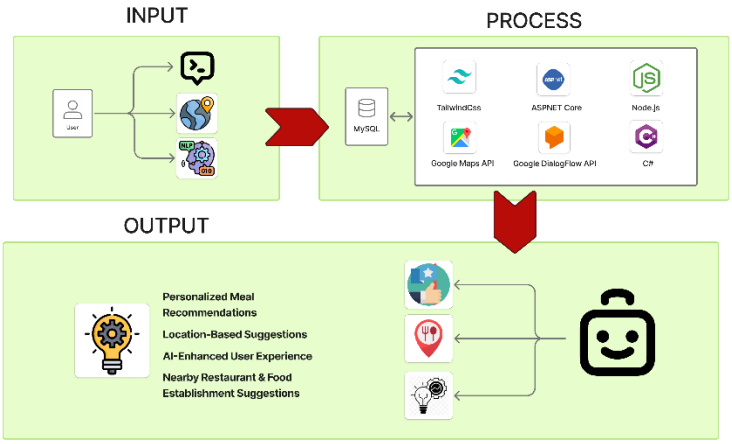


Figure 2 Conceptual Framework

This Framework follows the Input-Process-Output (IPO) model which shows how all the components work together to deliver accurate and relevant meal suggestions.

In the input stage, users provide the necessary data including their food preferences, dietary restrictions and real-time location. They interact with the system through natural language queries asking for meal options that fit their needs. The system captures user preferences through profile creation, dietary restriction inputs and real-time location data via GPS coordinates. Users can specify their dietary needs (vegetarian, vegan, gluten-free etc.), budget constraints, preferred cuisine types and meal timing preferences. This input stage is the foundation for personalized recommendation generation.

The chatbot takes the user's input and uses a combination of technologies. At the NLP level, Google Dialogflow processes user queries for intent mapping, recommendation, and suggestion and the prescriptive recommendation system models the preference information for recommendation refinements. Geolocation, based on Google Maps API, locates the closest points of service according to user's food preferences. To correctly determine what businesses are nearby the user, the system utilizes the Haversine formula which is a calculation used to find the shortest distance between two points on a sphere given their latitudes and longitudes. This allows users to fine tune filtering of places by proximity, providing only those that lie within a user-specified distance. It even has a feedback loop, as user grades and reviews also inform subsequent suggestions. Data organization in SQL Server ensures a well-structured of meal options based on dietary needs and availability. Moreover, theory-practice application ensures that the chatbot follows research-based methodologies, maintaining credibility and effectiveness.

In the output stage, the chatbot delivers personalized meal recommendations based on individual preferences and restrictions. Users get location-based suggestions with practical dining options from nearby establishments. The system also makes meal selection faster and more efficient. A key feature of the Personalized Location-Based Food Assistant is its prescriptive recommendation system that allows it to suggest relevant options based on user interactions. Research-guided development ensures the system evolves with structured methodologies so it gets more accurate and effective. Feedback is included so the system can store and use user preferences and make meal recommendations more accurate and relevant over time.

2.3.2 Natural Language Processing Integration

A key part of the study is Natural Language Processing (NLP) a subfield of Artificial Intelligence (AI) which enables context aware user interactions. The system uses Google Dialogflow API as its conversational AI engine, allowing users to communicate their meal preferences, dietary restrictions, and location-based needs through natural language queries.

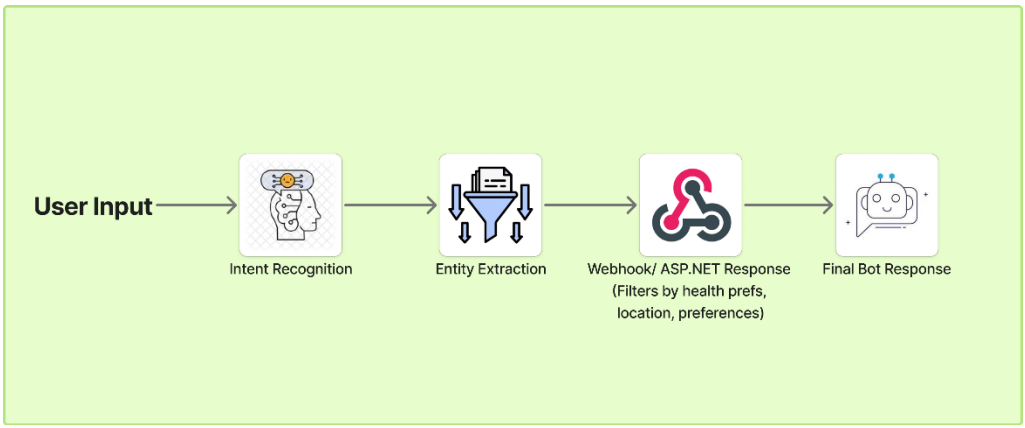


Figure 3 NLP workflow

Dialogflow's NLP pipeline enhances personalization by detecting the intent and parameters within user input, passing the extracted information to the backend server, where it is processed by the recommendation engine. This process can be summarized as:

1. The user interacts with the chatbot,
2. Dialogflow detects the intent and extracts parameters,
3. The API calls the server with extracted data,
4. The recommender processes the request,
5. and the system returns a contextually relevant response.

Prescriptive Recommendation Engine and Behavioral Learning

The system's recommendation algorithm looks at user profiles, past interactions and feedback to generate meal suggestions based on individual dietary needs. By storing and analyzing this data through Entity Framework Core and SQL Server the chatbot adapts its suggestions as user preferences change.

A major innovation is the prescriptive recommendation engine which goes beyond descriptive or predictive models. Unlike basic recommendation systems that only show similar meals or anticipate future preferences, the prescriptive engine actively suggests the best meal options based on user behavior, goals and constraints. It learns from user behavior by monitoring how users interact with the chatbot, what they choose and how they rate past recommendations. It analyzes these interactions to recognize consumption patterns and decision making and uses that to inform more goal oriented suggestions. Over time this engine will deliver more personalized and actionable recommendations that support healthier and time efficient decisions. To further increase the credibility of the system and reduce AI generated hallucinations web scraping is being considered to extract real time data from verified restaurant and food sources so recommendations are accurate

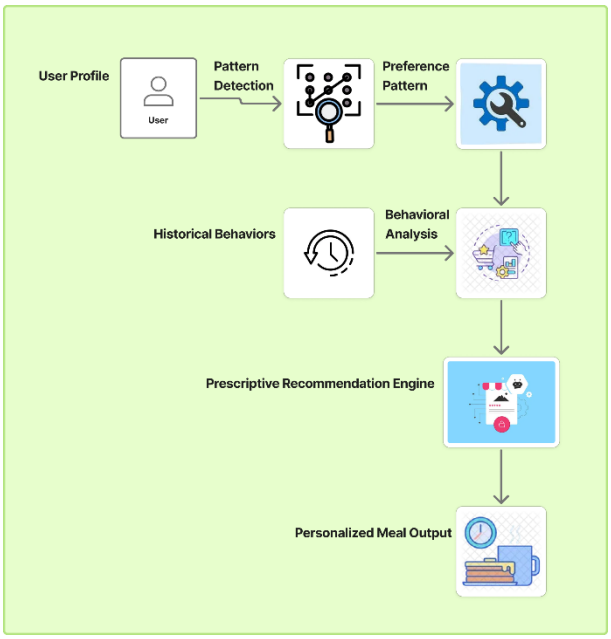


Figure 5 Prescriptive Recommendation Engine Workflow

The workflow above shows how user inputs, behavioral analysis, and the prescriptive engine work together to deliver relevant meal recommendations. This approach enables the system to learn and improve over time, supporting users in making informed and goal-oriented meal decisions.

Parameter Analysis and Recommendation Formula Implementation

The prescriptive recommendation engine uses a weighted scoring algorithm to generate optimal meal suggestions. Each parameter is systematically calculated based on data stored in the system's database, ensuring recommendations are derived from quantifiable user behaviors and preferences.

Recommendation Score Formula

The comprehensive recommendation score is computed as a weighted linear combination of five key parameters:

$$\begin{aligned}\text{Total Score} = {} & (\text{Dietary Restriction Weight} \times 0.25) + \\ & (\text{User Preference Weight} \times 0.30) + \\ & (\text{Nutritional Optimization Weight} \times 0.20) + \\ & (\text{Health Condition Weight} \times 0.10) + \\ & (\text{Behavioral Pattern Weight} \times 0.15)\end{aligned}$$

Figure 6 Recommendation Score Formula

This formulation prioritizes safety and health considerations while maintaining personalization through user preferences and behavioral patterns.

Parameter Calculation Methods:

1. Dietary Restrictions Weight (25% - High Priority)

Data Source: UserDietaryRestrictions table containing user-specific dietary constraints with importance levels (1-10 scale).

Mathematical Formulation:

For a user with n dietary restrictions, the dietary restriction compliance weight is calculated as:

If no restrtictions:

$$W_{dietary} = 1.0$$

Figure 7 Dietary Restrictions Weight if no restrictions

Otherwise:

$$W_{dietary} = \frac{\sum_{i=1}^{n}(C_i \times w_i)}{\sum_{i=1}^{n} w_i}$$

Figure 8 Dietary Restrictions Weight

Where:

- $C_i = 0$ or 1 (compliance score for restriction i)
- $w_i = \frac{\text{ImportanceLevel}_i}{10}$ (importance weight normalized to 0.1–1.0)

Figure 9

Compliance Assessment: Binary scoring (0.0 for violation, 1.0 for compliance) based on food content analysis:

- **Vegan:** No animal products (meat, dairy, eggs)
- **Vegetarian:** No meat products
- **Gluten-free:** No wheat, barley, rye derivatives
- **Dairy-free:** No milk-based ingredients
- **Nut-free:** No tree nuts or peanuts

2. User Preference Weight (30% - High Importance)

Data Source: FoodPreferences table storing user-selected preferred food types.

Mathematical Formulation:

$$W_{preference} = \begin{cases} 0.9 & \text{if food type is preferred} \\ 0.4 & \text{otherwise} \end{cases}$$

Figure 10 User Preferences Weight

Weight Assignment:

- Preferred food types: 0.9 (90% preference alignment)
- Non-preferred food types: 0.4 (40% neutral baseline)

3. Nutritional Optimization Weight (20% - Important) Data Source: FoodType nutritional properties analyzed through keyword matching and nutritional databases.

Mathematical Formulation:

$$W_{nutrition} = \min\left(1.0,\ \max(0.0,\ S_{base} + S_{quality} + S_{micronutrient} + S_{health-specific})\right)$$

Where:

- $S_{base} = 0.5$
- $S_{quality} = (0.2 \times I_{healthy}) + (0.1 \times I_{low-cal} \times Q_{healthy}) + (0.1 \times I_{protein})$
- $S_{micronutrient} = (0.15 \times I_{nutrient-dense}) + (0.15 \times I_{vitamin-rich})$
- $S_{health-specific} = \sum_{j=1}^{m}(0.1 \times A_j \times \frac{severity_j}{10})$

(Where I_x are binary indicators, and A_j is alignment with condition j.)

Figure 11Nutritional Optimization Weight

Assessment Categories:

- General nutritional quality: Healthy food indicators, caloric density, protein content
- Micronutrient density: Vitamin and mineral richness
- Health-specific alignment: Condition-appropriate nutritional profiles

4. Health Condition Weight (10% - Medical Safety)

Data Source: UserHealthConditions table with severity levels (1-10 scale).

Mathematical Formulation:

For a user with health conditions, the health condition alignment weight is calculated as:

If no health conditions:

$$W_{health} = 1.0$$

Otherwise:

$$W_{health} = \frac{\sum_{j=1}^{m}(A_j \times s_j)}{\sum_{j=1}^{m} s_j}$$

Where:

- A_j = alignment score (0.3 – 1.0)
- $s_j = \frac{\text{SeverityLevel}_j}{10}$ (normalized severity)

Figure 12 Health Condition Weight

Condition-Specific Alignments:

- Diabetes: Low glycemic index foods, sugar content assessment

- Hypertension: Low sodium, heart-healthy options

- Heart disease: Cardiovascular-friendly nutritional profiles

- Obesity: Low-calorie, nutrient-dense options

- Anemia: Iron-rich food sources

Mathematical Formulation:

5. Behavioral Pattern Weight (15% - Personalization)

Data Source:** UserPreferencePatterns table containing learned patterns from user interactions.

Sub-components (Equal Distribution):

- Temporal Patterns (5%): Time-based preference analysis

- Contextual Patterns (5%): Location and situational preferences

- Historical Patterns (5%): Cuisine and food type interaction history

$$W_{behavioral} = \min\left(1.0,\ S_{base} + S_{cuisine} + S_{temporal} + S_{contextual}\right)$$

Where:

- $S_{base} = 0.5$
- $S_{cuisine} = 0.3 \times \text{Confidence}_{cuisine} \times M_{cuisine}$
- $S_{temporal} = 0.2 \times \text{Confidence}_{time} \times M_{time}$
- $S_{contextual} = 0.2 \times \text{Confidence}_{location} \times M_{location}$

Confidence:

$$\text{Confidence} = \min\left(1.0,\ \frac{\text{Pattern Occurrences}}{\text{Total Interactions}}\right)$$

Figure 13 Behavioral Pattern Weight

Sample Datasets

1. User Profile Dataset

User Preferences

Table 3 User Preferences

UserID	Age	Location	PreferredCuisines	DietaryRestrictions	Health Conditions
U001	28	Downtown	Asian, Italian	Vegetarian	None
U002	35	University	Mediterranean. Thai	Vegan, Low-sodium	Type2 Diabetes, Hypertension
U003	42	Mall Area	Korean, Japanese	Gluten-free	Heart Disease

User Dietary Restrictions

Table 4 User Dietary Restrictions

User ID	Dietary Restrictions	ImportanceLevel	Compliance Score
U001	Vegetarian	9	1.0
U002	Vegan	10	1.0
U002	Low-sodium	7	0.8
U003	Gluten-free	8	1.0

User Health Conditions

Table 5 User Health Conditions

User ID	HealthConditions	SeverityLevel	AlignmentScore
U001	None	-	1.0
U002	Type2 Diabetes	7	0.9
U003	Hypertension	5	0.8
U004	Heart Disease	6	0.9

2. Food Type Analysis Dataset

Table 6 Food Type Analysis Dataset

FoodTypeID	FoodName	CuisineType	IsHealthy	IsLowCal	IsProtein	IsNutrientDense	IsVitaminRich
F001	Vegetable Stir-fry	Asian	1	1	0	1	1
F002	Quinoa Mediterranean Bowl	Mediterranean	1	1	1	1	1
F003	Grilled Salmon Teriyaki	Japanese	1	0	1	1	0
F004	Vegan Thai Curry	Thai	1	1	1	1	1
F005	Korean Bibimbap	Korean	1	0	1	1	1

3. Behavioral Pattern Dataset

Table 7 Behavioral Pattern Dataset

UserID	PatternType	PatternValue	Confidence	Occurrences	TotalInteractions
U001	cuisine	Asian	0.85	17	20
U001	time	evening	0.70	14	20
U001	location	downtown	0.65	13	20
U002	cuisine	Mediterranean	0.90	18	20
U002	time	afternoon	0.60	12	20
U002	location	university	0.75	15	20
U003	cuisine	Korean	0.75	15	20
U003	time	evening	0.80	16	20
U003	location	mall	0.70	14	20

Complete Parameter Substitution Example

Scenario 1: User U001 Analysis

User Profile: UserID U001, age 28, vegetarian preference

Query: "healthy Asian dinner near me" at 7:00 PM

Location: Davao City downtown area

Food Option 1: “Vegetable Stir-fry” (FoodTypeId: F001)

Parameter Calculation Process:

1. Dietary Restriction Weight Calculation:

$w_1 = 9/10 = 0.9$

W_dietary = (1.0 × 0.9) / 0.9 = 1.0

2. User Preference Weight Calculation:

Asian cuisine in preferred list: True

W_preference = 0.9

3. Nutritional Optimization Weight Calculation:

S_base = 0.5

S_quality = 0.2(1) + 0.1(1)(1) + 0.1(0) = 0.3

S_micronutrient = 0.15(1) + 0.15(1) = 0.3

S_health-specific = 0 (no health conditions)

W_nutrition = min(1.0, 0.5 + 0.3 + 0.3 + 0) = 1.0

4. Health Condition Weight Calculation:

No health conditions exist

W_health = 1.0

5. Behavioral Pattern Weight Calculation:

S_base = 0.5

S_cuisine = 0.3 × 0.85 × 1 = 0.255

S_temporal = 0.2 × 0.70 × 1 = 0.14

S_contextual = 0.2 × 0.65 × 1 = 0.13

W_behavioral = min(1.0, 0.5 + 0.255 + 0.14 + 0.13) = 1.0

Final Score Calculation:

Total Score = (1.0 × 0.25) + (0.9 × 0.30) + (1.0 × 0.20) + (1.0 × 0.10) + (1.0 × 0.15)

Total Score = 0.25 + 0.27 + 0.20 + 0.10 + 0.15

Total Score = 0.97

Food Option 2: "Grilled Salmon Teriyaki" (FoodTypeId: F003)

Parameter Calculation Process:

1. Dietary Restriction Weight Calculation:

Vegetarian compliance: 0.0 (contains fish)

$w_1 = 9/10 = 0.9$

W_dietary = (0.0 × 0.9) / 0.9 = 0.0

2. User Preference Weight Calculation:

Japanese cuisine in preferred list: True (Asian preference)

W_preference = 0.9

3. Nutritional Optimization Weight Calculation:

S_base = 0.5

S_quality = 0.2(1) + 0.1(0)(1) + 0.1(1) = 0.3

S_micronutrient = 0.15(1) + 0.15(0) = 0.15

S_health-specific = 0 (no health conditions)

W_nutrition = min(1.0, 0.5 + 0.3 + 0.15 + 0) = 0.95

4. Health Condition Weight Calculation:**

No health conditions exist

W_health = 1.0

5. Behavioral Pattern Weight Calculation:

S_base = 0.5

S_cuisine = 0.3 × 0.85 × 1 = 0.255

S_temporal = 0.2 × 0.70 × 1 = 0.14

S_contextual = 0.2 × 0.65 × 1 = 0.13

W_behavioral = min(1.0, 0.5 + 0.255 + 0.14 + 0.13) = 1.0

Final Score Calculation:**

Total Score = (0.0 × 0.25) + (0.9 × 0.30) + (0.95 × 0.20) + (1.0 × 0.10) + (1.0 × 0.15)

Total Score = 0.0 + 0.27 + 0.19 + 0.10 + 0.15 = 0.71

Food Option 3: "Korean Bibimbap" (FoodTypeId: F005)

Parameter Calculation Process:

1. Dietary Restriction Weight Calculation:

Vegetarian compliance: 1.0 (vegetable-based bibimbap)

w_1 = 9/10 = 0.9

W_dietary = (1.0 × 0.9) / 0.9 = 1.0

2. User Preference Weight Calculation:

Korean cuisine in preferred list: False (not in Asian preference)

W_preference = 0.4

3. Nutritional Optimization Weight Calculation:**

S_base = 0.5

S_quality = 0.2(1) + 0.1(0)(1) + 0.1(1) = 0.3

S_micronutrient = 0.15(1) + 0.15(1) = 0.3

S_health-specific = 0 (no health conditions)

W_nutrition = min(1.0, 0.5 + 0.3 + 0.3 + 0) = 1.0

4. Health Condition Weight Calculation:

No health conditions exist

W_health = 1.0

5. Behavioral Pattern Weight Calculation:**

S_base = 0.5

S_cuisine = 0.3 × 0.85 × 0 = 0.0 (Korean ≠ Asian pattern)

S_temporal = 0.2 × 0.70 × 1 = 0.14

S_contextual = 0.2 × 0.65 × 1 = 0.13

W_behavioral = min(1.0, 0.5 + 0.0 + 0.14 + 0.13) = 0.77

Final Score Calculation:

Total Score = (1.0 × 0.25) + (0.4 × 0.30) + (1.0 × 0.20) + (1.0 × 0.10) + (0.77 × 0.15)

Total Score = 0.25 + 0.12 + 0.20 + 0.10 + 0.116 = 0.78

Scenario 1 Ranking Results:

Food Option	Total Score	Rank	Safety Status	Recommendation Status
Vegetable Stir-fry	0.97	1	Safe (dietary compliance = 1.0)	Highly Recommended
Korean Bibimbap	0.786	2	Safe (dietary compliance = 1.0)	Recommended
Grilled Salmon Teriyaki	0.71	3	**Unsafe** (dietary compliance = 0.0)	**Not** Recommended

U001's results show us some interesting things about the algorithm. Vegetable Stir-fry is the clear winner with a score of 0.97, maxing out on dietary restrictions, nutritional quality and behavior while

matching the user's documented Asian cuisine preference. Korean Bibimbap is vegetarian safe but scored 0.786, because Korean cuisine is outside the user's recorded Asian preference profile so the preference weighting is lower (0.4 vs 0.9) and the behavior correlation is lower. The safety mechanism worked when Grilled Salmon Teriyaki scored 0.71 – despite strong preference alignment and behavior consistency, the complete dietary restriction violation (0.0 compliance) shows that the 25% safety weighting prevents recommendations that can harm users, no matter how good the other parameters are.

Geolocation Integration and Haversine Distance Calculation

Another important element is the incorporation of geolocation services to provide contextually relevant meal suggestions. The system uses the Google Maps API to obtain real-time user location data, which is then used to identify nearby food establishments. To determine which establishments are within the user's vicinity, the system implements the Haversine formula, a mathematical algorithm for calculating the shortest distance between two points on the Earth's surface using latitude and longitude coordinates. The Haversine formula is also the core distance computation behind Google Maps' Route API and similar geospatial systems. By this formula the system filters and recommends only restaurants within a certain radius, makes suggestions more relevant and accessible. The formula is as follows:

$$d = 2r \cdot \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2\left(\frac{\Delta\lambda}{2}\right)}\right)$$

where:

- r is Earth's radius (≈ 6,371 km),
- $\Delta\phi$ and $\Delta\lambda$ are differences in latitude and longitude (in radians),
- ϕ_1 and ϕ_2 are the latitudes of the two points.

Figure 4 Haversine Formula

Where:

- r is Earth's radius (≈ 6,371 km),
- $\Delta\phi$ and $\Delta\lambda$ are the differences in latitude and longitude (in radians),
- $\phi 1$ and $\phi 2$ are the latitudes of the two points.

By this formula the system recommends only restaurants within a certain radius, makes suggestions more practical.

Responsive Web Interface for User Interaction

The responsive web interface is another major component, designed using Tailwind CSS to support accessibility and usability across a range of devices. The interface allows users to register, input preferences, interact with the chatbot, and view recommendations. By prioritizing clarity and responsiveness, the system supports both desktop and mobile access for meal planning and food discovery.

2.3.3 Data Models

The researchers will show the data management of the application through data models, the structure and organization of the data. Use-case diagrams will show user interactions and functionalities, entity relationship diagrams will show the relationships between the different data entities in the database. This will give a full view of the system design, architecture and functionality and how it handles and processes data.

2.3.3.1 Use Case Diagrams

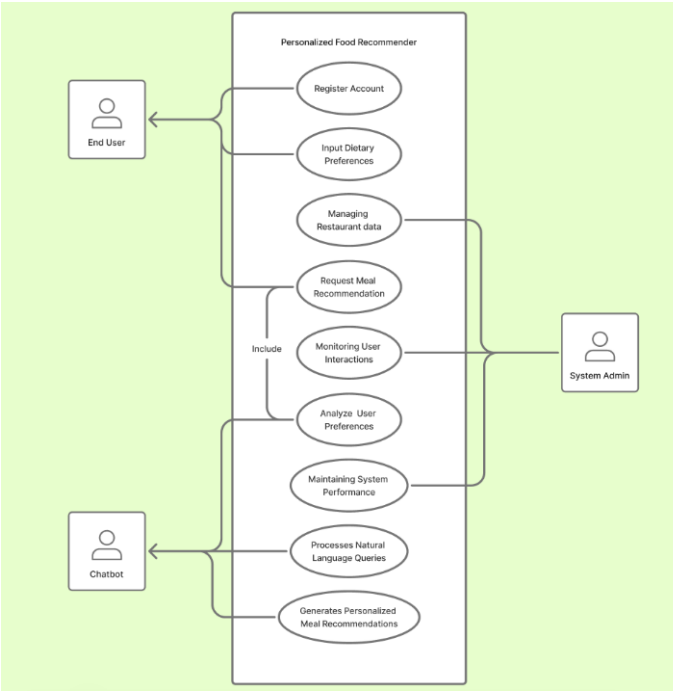


Figure 8 Use Case Diagram

Based on the Use Case diagram, it shows 3 main users: end users (customers), system administrators and the AI chatbot system. End users interact with the system to register, input dietary preferences, request meal recommendations and provide feedback on suggestions. System administrators manage the system by managing restaurant data, monitoring user interactions and maintaining system performance. The AI chatbot system processes natural language queries, analyze user preferences and generate meal recommendations. For location-based queries, the chatbot uses the Haversine formula to filter and recommend only the restaurants that are within the user's proximity as determined by their real-time coordinates. This diagram shows how these roles are connected and how important each is in having an ideal and efficient food recommendation experience.

Figure 9 Entity Relationship Diagram

The Entity-Relationship Diagram (ERD) illustrates how all our data, or entities, are connected in our Microsoft SQL Server database. It provides us a view on the database scheme, on how information is stored and how it is related. The diagram shows the

relationships between User, UserProfile, FoodType, FoodPreference, ChatSession, ChatMessage, Establishment entities, how the data flows from user interactions to recommendation generation. The ERD also shows how we store latitude and longitude coordinates for both users and establishments which is used for the Haversine formula based distance calculations that powers the location based recommendation feature.

2.3.3.3 Data Dictionary

Table 3 Users Table

Field Names	Datatype	Length	Description
UserID-PK	Int-AI	9	User's unique identifier
Name	Text	100	User's full name
Email	Text	100	User's unique email address
PasswordHash	Text	100	User's encrypted password
CreatedAt	Text (Date)	25	Timestamp of account creation
LastLogin	Text (Date)	25	Timestamp of the user's last login

The Users table stores all the basic information about the people who use our system. It keeps their login details and profile information safe which we need to personalize their experience and keep the application secure.

Table 4 UserProfiles Table

Field Names	Datatype	Length	Description
ProfileID-PK	Int-AI	9	User profile's unique identifier
UserID-FK	Int	9	Foreign key referencing the Users table
DietaryPreference	Text	50	User's specified dietary restrictions
FavoriteFood	Text	100	User's listed favorite food
PreferredLocation	Text	200	User's preferred area for dining
PriceRange	Text	20	User's preferred price range
LastKnownLatitude	Decimal	10	Last known latitude of the user
LastKnownLongitude	Decimal	11	Last known longitude of the user
LastUpdated	Text (Date)	25	Timestamp of the last profile update

The UserProfiles table stores demographic and physiological data for each user. These fields are used in conjunction with establishment coordinates to calculate proximity using the Haversine formula, enabling precise location-based filtering of recommendations.

Table 5 FoodTypes Table

Field Names	Datatype	Length	Description
FoodTypeID-PK	Int-AI	9	Food category's unique identifier
Name	Text	50	Name of the food category (e.g., Italian)
Description	Text	200	Brief description of the food category

The FoodTypes table functions as a centralized, controlled vocabulary for all food and cuisine classifications used within the application. Its purpose is to standardize the nomenclature for categories such as "Italian," "Vegan," or "Dessert." This normalization is essential for data integrity and consistency, ensuring that user queries, restaurant menu items, and learned preference patterns are all mapped to a uniform set of descriptors. This standardization is a prerequisite for the effective functioning of the matching and recommendation algorithms.

Table 6 FoodPreferences Table

Field Names	Datatype	Length	Description
PreferenceID-PK	Int-AI	9	Preference record's unique identifier
UserID-FK	Int	9	Foreign key referencing the Users table
FoodTypeID-FK	Int	9	Foreign key referencing the FoodTypes table
PreferenceLevel	Text	20	User's level of preference (e.g., High, Medium)
LastSelected	Text (Date)	25	Timestamp when this preference was last selected

This table explicitly maps users to their self-declared food preferences, creating a many-to-many relationship between the Users and FoodTypes tables. The purpose of this table is to persist the user's initial and manually configured taste profile. This data serves as the baseline for the personalization algorithm, providing a starting point for recommendations before the system has accumulated sufficient behavioral data to infer implicit preferences.

Table 7 ChatSessions Table

Field Names	Datatype	Length	Description
SessionID-PK	Int-AI	9	Chat session's unique identifier
UserID-FK	Int	9	Foreign key referencing the Users table
StartedAt	Text (Date)	25	Timestamp when the chat session started
EndedAt	Text (Date)	25	Timestamp when the chat session ended

The ChatSessions table is designed to manage and contextualize user interactions. Each record represents a single, distinct conversational instance between a user and the chatbot. Its purpose is to track the lifecycle of an interaction, including start and end times, thereby enabling the system to maintain conversational state. This segmentation is critical for subsequent analysis, as it allows user behavior to be examined within the context of a specific goal or session.

Table 8 ChatMessages Table

Field Names	Datatype	Length	Description
MessageID-PK	Int-AI	9	Chat message's unique identifier
SessionID-FK	Int	9	Foreign key referencing the ChatSessions table
Sender	Text	20	The sender of the message (User or AI)
MessageText	Text	1000	The content of the chat message
Timestamp	Text (Date)	25	Timestamp when the message was sent
IntentDetected	Text	50	The intent recognized by Dialogflow

The ChatMessages table serves as an archive of the dialogue between the user and the system. It meticulously logs every user-generated query and every AI-generated response, linking them to a specific ChatSession.

Table 9 UserBehaviors Table

Field Names	Datatype	Length	Description
BehaviorID-PK	Int-AI	9	Behavior record's unique identifier
UserID-FK	Int	9	Foreign key referencing the Users table
Action	Text	50	The action performed by the user (e.g., search)
Context	Text	200	The context of the action (e.g., query text)
Result	Text	500	The result of the action (e.g., intent name)
Timestamp	Text (Date)	25	Timestamp when the behavior occurred
Satisfaction	Int	1	User's satisfaction rating (1-5)
IntentDetected	Text	100	The Dialogflow intent was detected for the behavior
ParametersJson	Text	2000	Parameters extracted for the behavior (JSON)

This table is the primary data sink for the system's implicit feedback loop. Its purpose is to log discrete, meaningful user actions that signify preference or intent. These actions include search queries, selection of a specific recommendation, interaction with certain UI elements (e.g., ratings, saving an item), and the context (e.g., time of day, location) in which these actions occurred. Each record in this table is a behavioral signal that serves as a direct input for the prescriptive recommendation engine.

Table 10 RecommendationHistories Table

Field Names	Datatype	Length	Description
HistoryID-PK	Int-AI	9	History record's unique identifier
UserID-FK	Int	9	Foreign key referencing the Users table
RecommendedAt	Text(Date)	25	Timestamp when the recommendation was made
SelectedAt	Text(Date)	25	Timestamp when the user selected it
UserRating	Int	1	User's rating for the recommendation (1-5)
ReasonFor Recommendation	Text	200	The logic behind why it was recommended
UserFeedback	Text	500	Text feedback provided by the user
EstablishmentInfo	Text	1000	JSON object with establishment details
EstablishmentName	Text	100	Name of the recommended establishment
CuisineType	Text	50	Cuisine type of the recommended establishment
PriceRange	Text	20	Price range of the recommended establishment

The RecommendationHistories table logs the output of the recommendation engine and the user's subsequent interaction with that output. Its purpose is to create a detailed record of which recommendations were presented to the user and whether those recommendations were acted upon (i.e., selected) or ignored.

Table 11 UserPreferencePatterns Table

Field Names	Datatype	Length	Description
PatternID-PK	Int-AI	9	Pattern record's unique identifier
UserID-FK	Int	9	Foreign key referencing the Users table
PatternType	Text	50	The type of pattern (time, location, cuisine, price)
PatternValue	Text	200	The value of the pattern (e.g., morning, downtown)
Confidence	Decimal	3	The confidence score for this pattern (0.00-1.00)
LastObserved	Text (Date)	25	Timestamp when the pattern was last observed
ObservationCount	Int	9	Total number of times this pattern was observed

This table represents the materialized output of the behavior learning module. Its purpose is to store the discovered, implicit preference patterns of a user, which are synthesized from the raw data in the UserBehaviors and RecommendationHistories tables. Each record quantifies a specific user preference (e.g., "prefers spicy food on weekdays") and is associated with a dynamic confidence score that reflects the strength of the statistical evidence for that pattern. This table directly drives the system's prescriptive recommendation capabilities, enabling it to generate proactive, highly personalized recommendations that adapt over time. Location-based patterns are derived from repeated proximity calculations using the Haversine formula, allowing the system to learn and adapt to users' preferred dining areas over time.

2.3.3.4 Architecture Design

The system uses ASP.NET Core and C# for backend processing, Google Dialogflow API for natural language understanding, and Google Maps API for geolocation services. Data is managed with Microsoft SQL Server, and Tailwind CSS provides a responsive user interface design. The prescriptive recommendation engine supports real-time recommendation generation, user preference learning, and behavioral pattern analysis. The system supports efficient data management through Entity Framework Core.

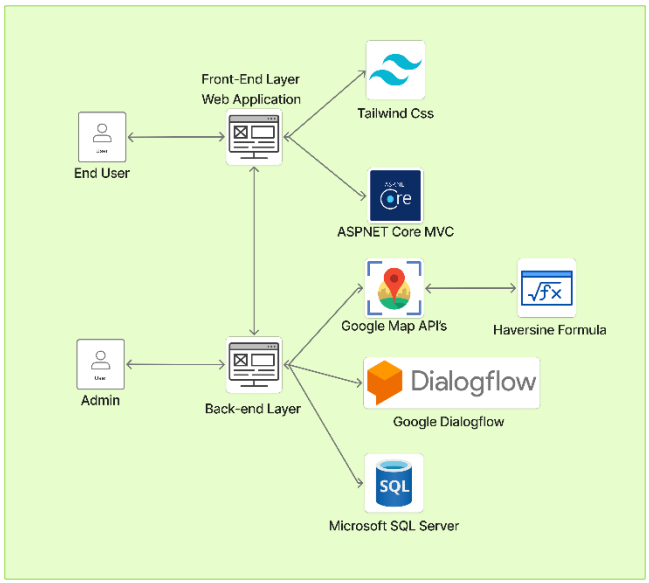


Figure 7 System Architecture

The system architecture diagram involves end users who can register accounts, input dietary preferences, request meal recommendations through natural language queries, and receive prescriptive suggestions based on their location and preferences. The AI chatbot processes user inputs using Google Dialogflow, analyzes preferences using custom algorithms, and generates prescriptive recommendations by querying the database for suitable establishments. System administrators manage restaurant data and monitor system performance. Since all users interact through web browsers, a centralized database stores and manages all user-related data, conversation history, and establishment information. A critical component of the system architecture is the integration of the Haversine formula within the backend logic. This enables the system to compute the distance between the user's current location and each food establishment, ensuring that only those within a specified radius are included in the recommendation set.

2.3.3.5 Sample Prototypes

2.3.3.5.1 Desktop Interface

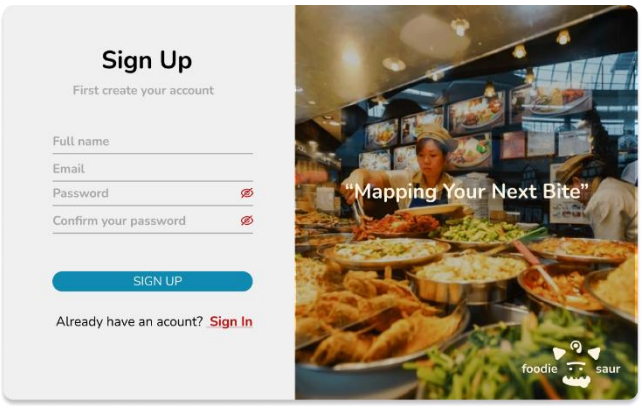


Figure 8 Sign-up Page

The sign-up screen displays input fields for full name, email, password, and confirm password for new users before they can access the main page. After completing the form, the user can click the sign-up button. If the user already has an account, they can choose to sign in instead.

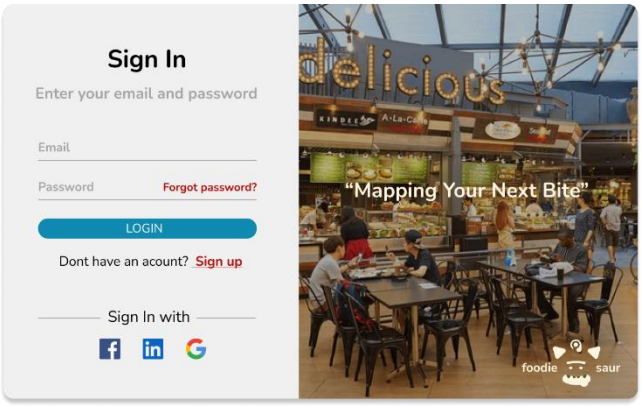


Figure 9 Login Page

This figure illustrates a mobile application login interface featuring two vertically positioned input fields for email and password, both clearly labeled for ease of use, with a "Forgot Password" link beneath the password field to help users recover their credentials if needed. Below these elements, a prominent Login button allows access to the main application, while alternative login options are offered through recognizable social media buttons for Facebook, Google, and LinkedIn, each branded with its respective colors and icons to support third-party authentication.

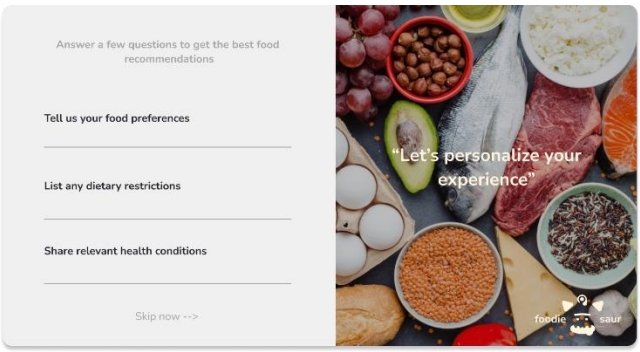


Figure 10 User Onboarding

The onboarding asks 3 questions about food preferences, health conditions and dietary restrictions to personalize the experience.

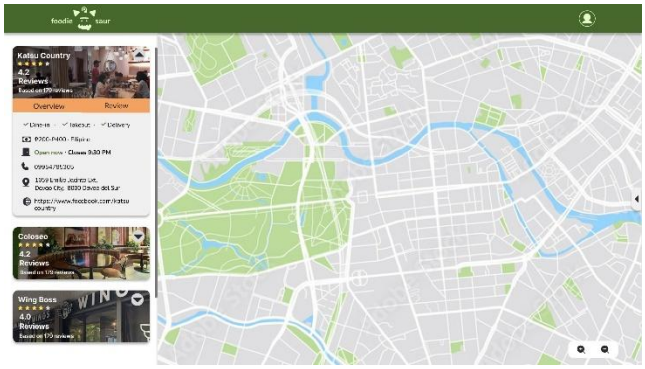


Figure 11 Main Page

The main page shows popular shops and a map to find each one, with a zoom tool to navigate closer. It also shows an overview and customer reviews of a shop, click the care button at the top right to reveal more.

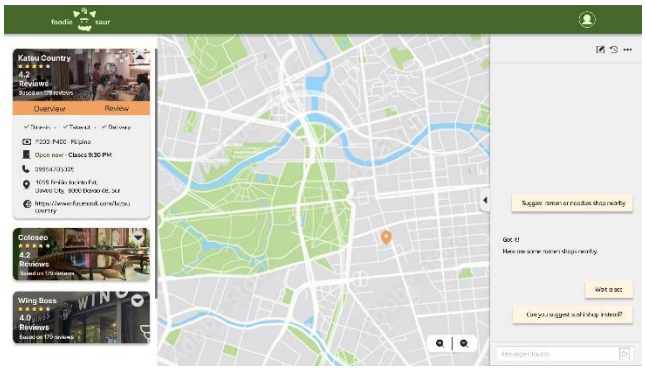


Figure 12 Chat Box

The chat box allows you to talk to the chatbot by asking or giving commands to get shop n recommendations. It has tools to start new conversations, view chat history and more options to get more details.

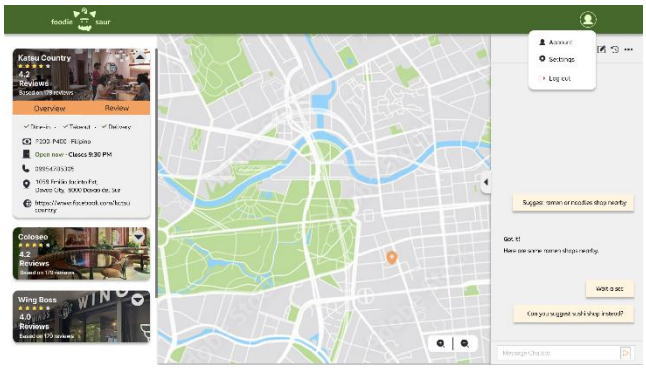


Figure 13 Log out

Clicking the user icon will take you to the user account, settings and logout options to exit the site.

2.3.3.5.2 Mobile Interface

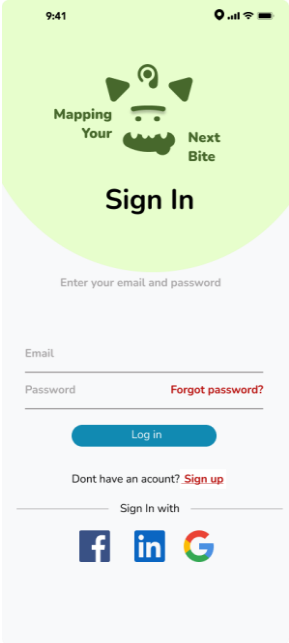


Figure 14 Sign in

The mobile sign-up screen has fields for full name, email, password and password confirmation. New users must fill in all fields before they can proceed. A visible "Sign Up" button to submit the form. For users who already have an account, there's an option to switch to the sign-in screen.

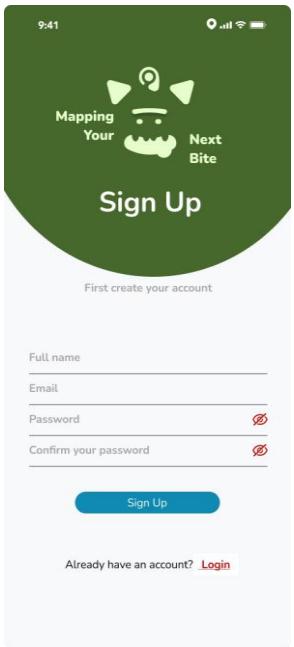


Figure 15 Sign up

To sign up, the mobile sign up page has fields for full name, email, password and password confirmation. If you already have an account you can sign in.

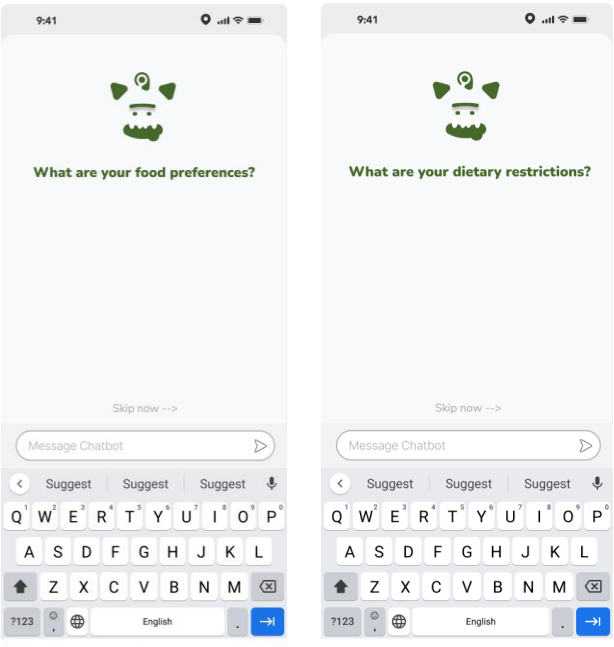


Figure 16 Onboarding

The personalization process starts with the mobile onboarding screen which has 3 targeted questions to gather information about food preferences, health conditions and dietary restrictions.

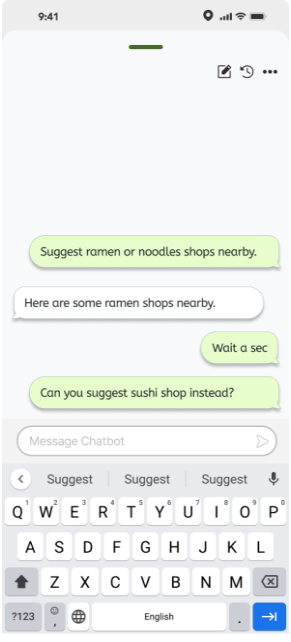


Figure 17 Mobile Chat Bot

Through the chat interface users can talk to the chatbot by asking questions or giving commands to get shop recommendations. Also integrated in this interface are tools to start new conversations, review chat history and access extra options for more functionality.

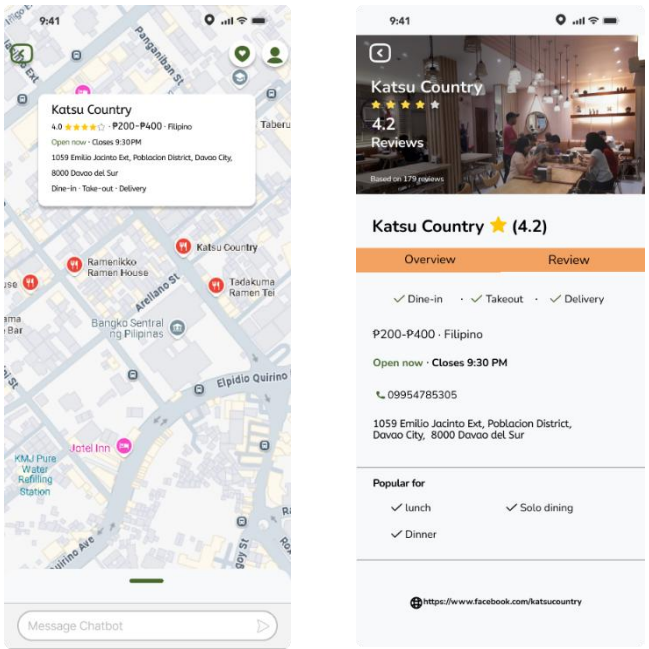


Figure 18 Overview

The core of the recommendation system is the overview screen which has a list of shops tailored to the individual user. Each recommendation has establishment name, imagery, description, user ratings, proximity data and operational information (opening hours and price range). If users need more information they can select any item to see reviews, extended description or navigation help.

2.4 Development

The proponents began planning by implementing project initiation and requirement gathering for the potential users to understand the user's needs. To make the chatbot functional the proponents implemented natural language processing and created a conversation flow to know the user's intent and to validate the chatbot's ability until it understands and responds accurately. The proponents implemented algorithms to build part of the proposed project that gives meal suggestions and studies each user's profile, preferences, and habits of what food they like. To have better experience in recommending food establishments, the proponents add a location feature to the system to measure how far nearby restaurants are and to provide an interface for navigation. The proponents improved the meal recommendations by applying linear score weight to measure how strongly a user prefers certain types of meals based on their choices and ratings. Finally, the deployment process in which the system will be available to users to perform a final test when it is accepted, the proponents proceed to complete the necessary documentation.

2.4.1 Project Initialization and Requirement Gathering

The proponents in planning and understanding the objectives of the project, defining what the proposed system would do, and gathering information from users through surveys to learn about their food preferences, dietary requirements, and need for a location-based feature. This helped the proponents create a list of features to guide the development process.

2.4.2 Chatbot and NLP API Development

The proponents implement integrating Google Dialogflow with the ASP.NET Core backend to enable natural language processing. This phase involved designing conversational flows, intents, and entities to ensure the chatbot could interpret user queries related to food preferences, health conditions, and location. Initial testing was conducted to validate the chatbot's ability to process and respond to user inputs accurately.

2.4.2.1 Design

The proponents used Tailwind CSS to provide a responsive UI for the desktop and mobile versions. The proponents decided to change the UI components to enhance their usability.

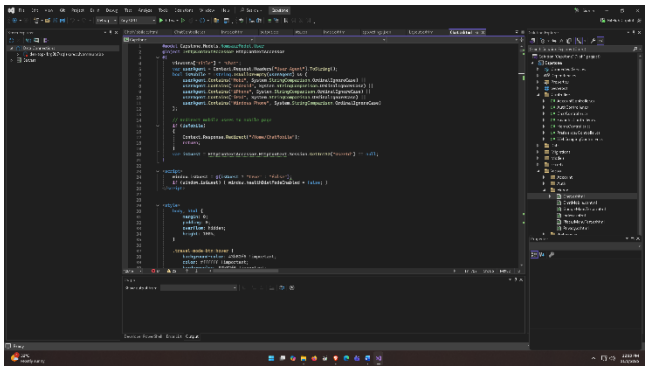


Figure 19 UI Design of the Website

The Figure shows above the UI design website development using Tailwind.css. This tab Chat.cshtml is where the main page is being created.

2.4.2.2 Code

The proponents programmed the authentication, buttons, and chatbot, and integrated the applied APIs to provide a better user experience while using the website.

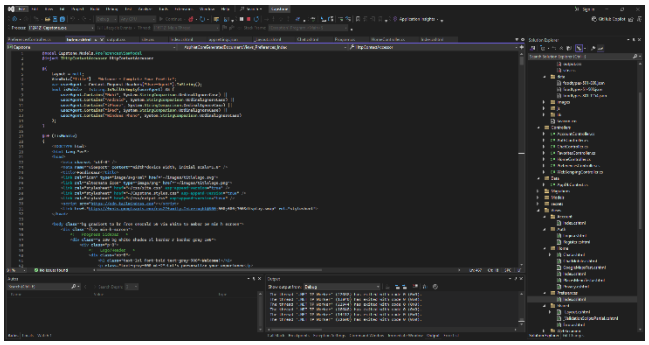


Figure 20 Code for the Sytem's Function

The figure below shows a section of the code for the website. The proponents used C# (ASP.NET Core MVC) to create the controllers for the website.

2.4.2.3 Updated Prototype

2.4.2.3.1 Desktop Interface

Figure 21 Landing Page

The Figure above shows the first interface, which displays information about the chatbot and its features. It also includes an interactive button to proceed to the next page.

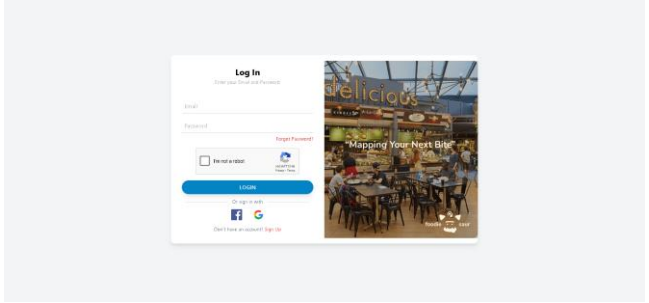


Figure 22 Log In

The figures show the Log in interface; the user can connect to the Facebook or google account for efficient logging in if the user does not have account this will proceed to the signing up page.

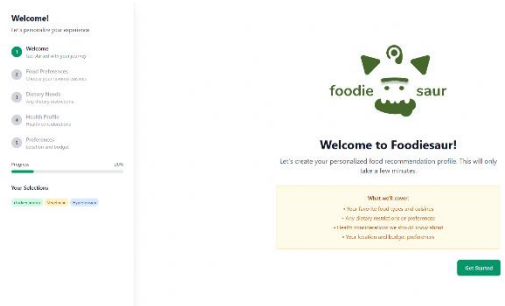


Figure 23 Onboarding

The figure above shows the setup for the user's preferences, dietary restrictions, and health conditions. This helps the chatbot provide accurate meal recommendations based on the user's profile. The user may also skip this setup if they prefer a more generalized query.

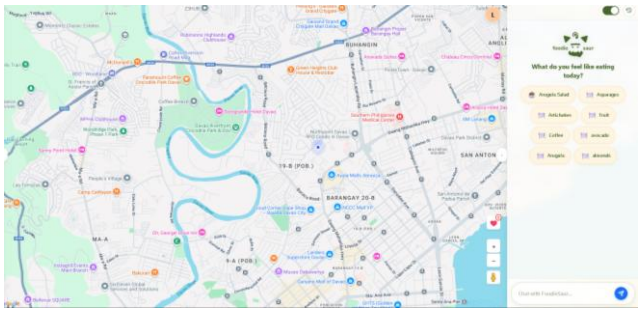


Figure 24 Main Page

This figure shows the chatbot interface, which includes suggestion buttons to assist users who are undecided. The heart icon allows users to save their favorite food establishments. The system also provides zoom in and zoom out controls for better map interaction, and a pegman icon when it dragged to a specific location will display the street view. At the upper right next to the switch icon is the chat history feature, which lets users review their previous conversations with the chatbot.

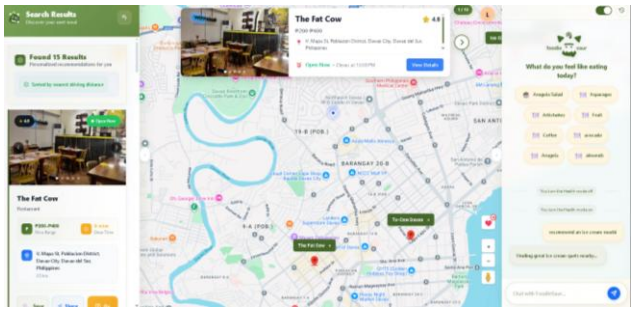


Figure 25 Meal Results

The figure shows that the feature displays the meal recommendation results, which include three buttons Save, Share, and Track Directions. The card in the middle presents detailed information about the meal, along with a View Details button. The user can choose the most suitable meal based on their preferences.

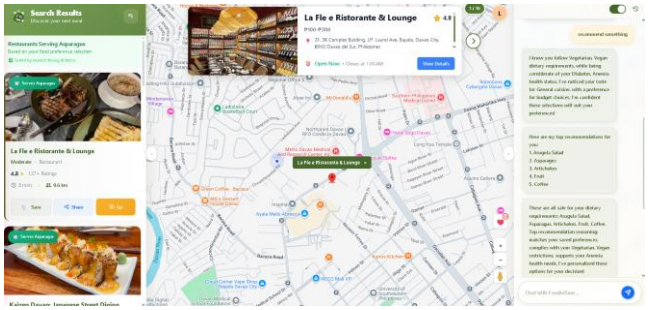


Figure 26 Health Mode on

The figures show the switch at the upper right indicates the mode when it turns green, the prescriptive query or health mode is activated, meaning the chatbot will recommend meals based on the user's preferences, dietary restrictions, and health conditions.

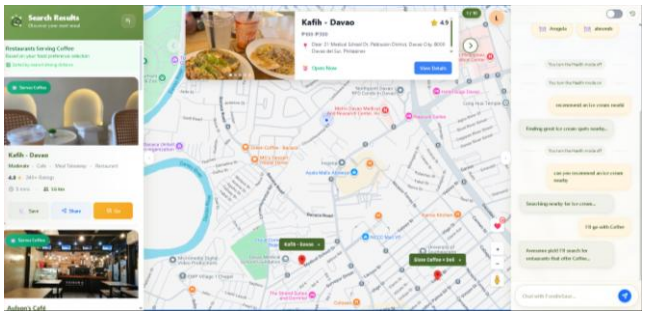


Figure 27 Health Mode off

The figure shows that when the switch is turned off, the chatbot provides generalized query recommendations that are not based on the user's personal data.

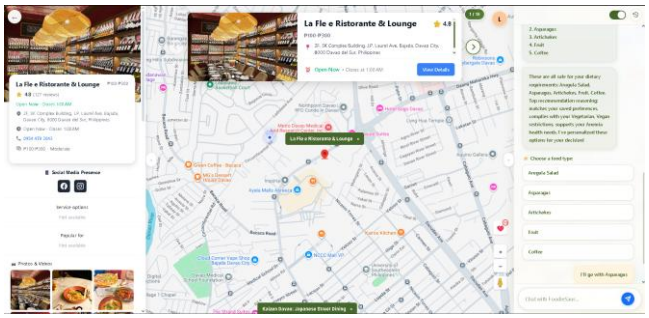


Figure 28 Overview

The figures show the details of the selected restaurant, including its location, price range, contact number, and operating hours. The system also provides a social media presence feature that allows users to view the restaurant's online platforms, along with available photos, highlights, menus, and reviews.

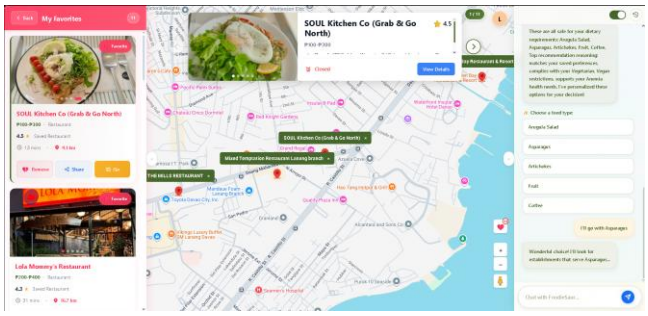


Figure 29 Favorites

The figure shows the user's saved food establishments and provides options to remove, share, or get directions for easier navigation.

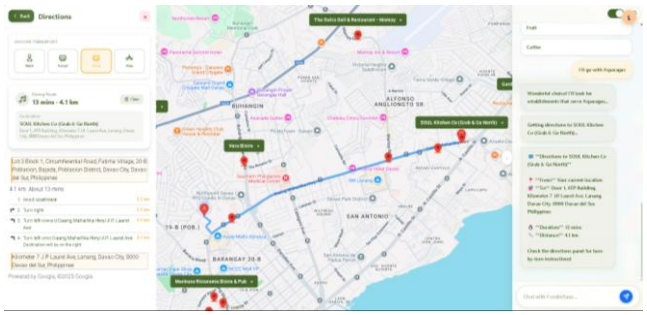


Figure 30 Direction Result

The figure shows the distance between the user and the restaurant, along with the direction panel. The user can choose a mode of transportation to measure the distance and travel duration, while the chatbot provides additional information about the location.

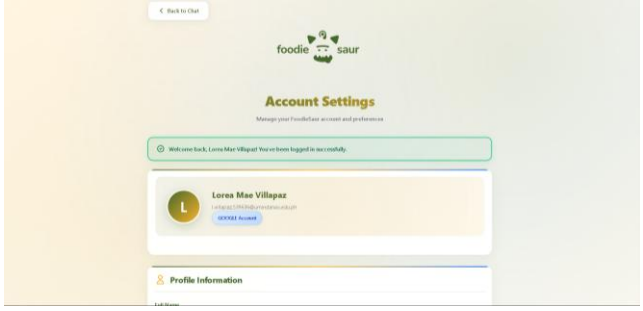


Figure 31 User Profile

The figure shows the account settings consist of profile information this allow the user to change name and save changes, along with edit preferences.

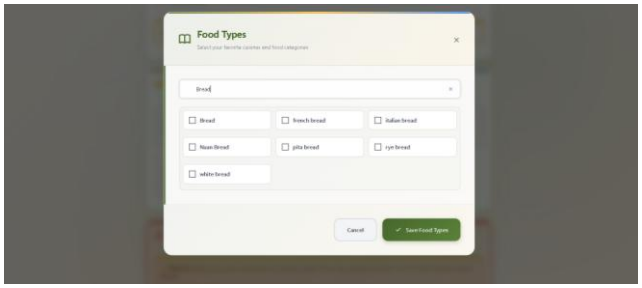


Figure 32 Edit Preferences

The figure shows that the user is allowed to change food preferences by selecting food types, dietary restrictions, and health conditions, as well as setting location and price ranges.

2.4.2.3.2 Mobile Interface

Figure 33 Landing Page

The figure shows the first mobile interface displaying the system introduction, where the user can click the button below to experience the chatbot.

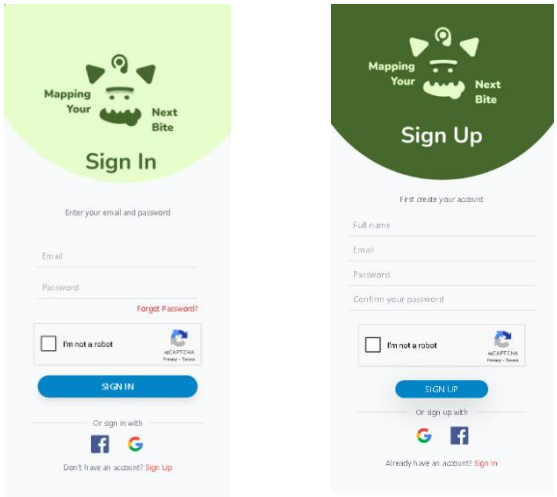


Figure 34 Log in and Sign In

This figure shows the Log in and sign in; the system has a reCAPTCHA feature for security. The user can connect using Facebook and Google account.

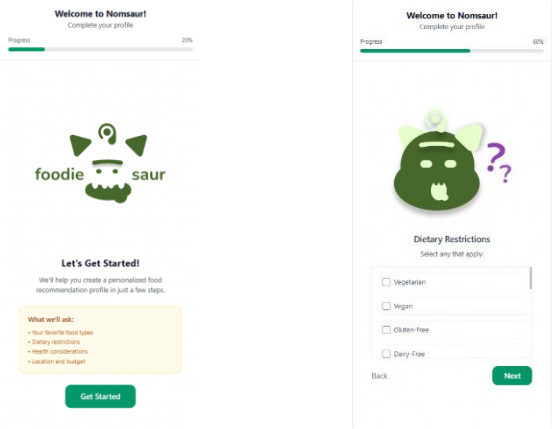


Figure 35 Onboarding

This figure shows the setting up for user's preferences consist of dietary restrictions and health conditions to have better experience for interacting with the system.

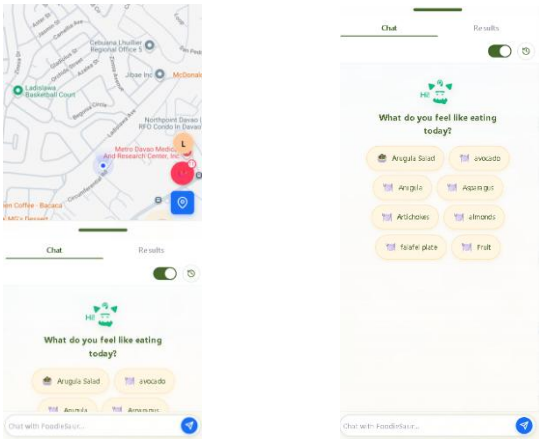


Figure 36 Main Page

This figure shows the chatbot and map for navigation the user allows to see results by clicking the result button, it also provides a various food type button inside the chatbot if ever the user is undecided.

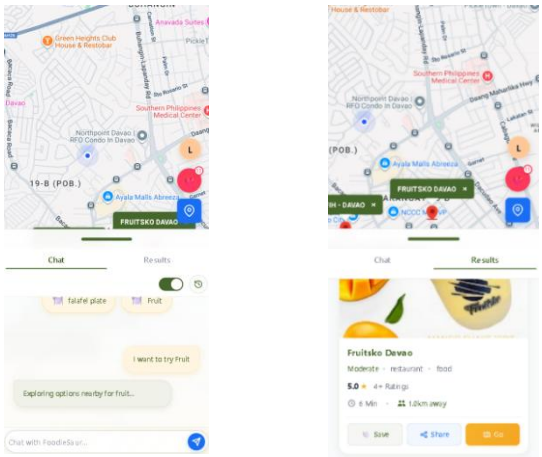


Figure 37 Meal Result

The figure shows the meal result and navigation to the chosen restaurant, where the recommendation is generated according to the user's preferences because the health mode is enabled.

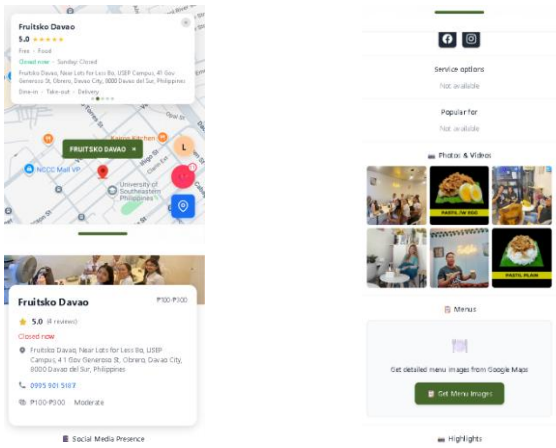


Figure 38 Overview

The figure shows the restaurant details with photos, highlights, and menu to provide more information about the restaurant.

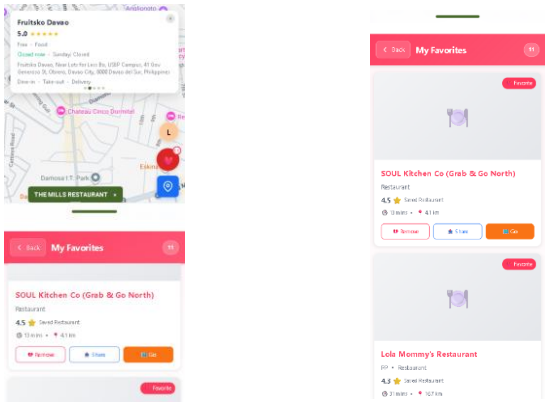


Figure 39 User Favorites

The figure shows the user's saved restaurants. This feature allows the user to access their favorite restaurants more easily and efficiently.

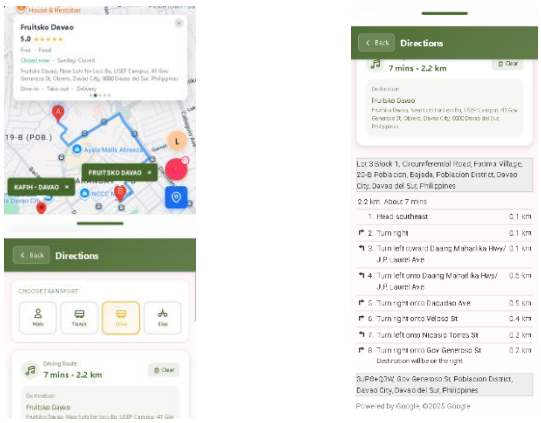


Figure 40 Direction Results

The figures show an interface with the restaurant directions so the user can choose transport to measure the distance and duration between user and the restaurant location.

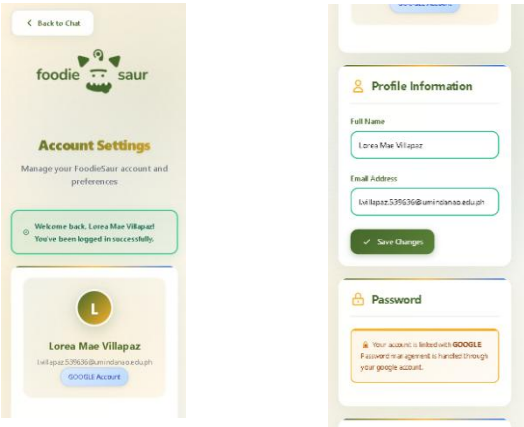


Figure 41 User Profile

The figure shows the user profile that allows user to edit the user information and password.

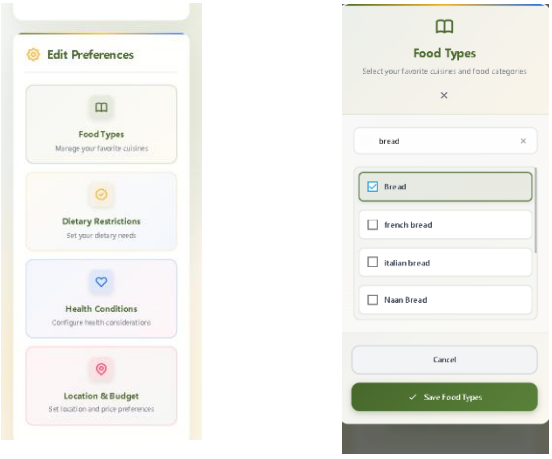


Figure 42 Edit Preferences

This figure shows that the user is allowed to change their preferences. It gives the user the freedom to choose the type of food they are interested in and to set their dietary needs, health conditions, preferred location, and price range.

2.4.3 AI and Personalization Integration

The proponents focused on developing the recommendation feature by implementing algorithms to analyze user profiles, preferences, and behavioral data. Using Entity Framework Core to manage user information stored in the SQL Server database, the system was configured with a prescriptive recommendation engine to generate meal suggestions tailored to each user's needs and preferences.

2.4.4 Geolocation Services Implemention

For better user interface when searching meals, the proponents developed the geolocation module by integrating the Google Maps API and implementing the Haversine formula for distance calculation. The proposed project was configured to detect the user's location, retrieve nearby food establishments, and filter recommendations based on proximity, with testing conducted to ensure that location-based suggestions were accurate and relevant.

2.4.5 Prescriptive Recommendation System Optimization

The proponents enhance the recommendation engine by using behavioral learning algorithms to study how users interact with the system and respond to suggestions. The system analyzed user feedback to make meal recommendations more accurate and relevant, and it was adjusted to automatically adapt to users' changing food preferences over time.

2.4.6 Final Deployment

This sprint aims on releasing the system to the production environment, performing final testing, and fixing any remaining issues. It also included completing all necessary documentation, making the system accessible to users, and setting up post-deployment monitoring to track system performance and ensure user satisfaction.

Table 12 Development Timebox

Sprint Activities	Duration	Responsible
Project Initiation and Requirement Gathering	2 weeks	Designer and Programmer
Chatbot and NLP API Development	3 weeks	Designer and Programmer
AI and Personalization Integration	2 weeks	Programmer
Geolocation Services Implementation	1 week	Designer and Programmer
Prescriptive Recommendation System Optimization	3 weeks	Programmer
Final Deployment phase	1 week	Designer and Programmer

2.5 Testing

The testing phase presents the evaluation results of the proposed project to ensure that the system functions accurately and runs smoothly. Several tests were conducted by the proponents, together with their adviser and target users, to assess the system's

performance and reliability. During alpha testing, the proponents and their adviser, Prof. Randy F. Ardeña, carefully examined the system to confirm that all objectives were met and every feature worked properly. Afterward, beta testing was carried out with participants, including the end user, to evaluate the system's efficiency, usability, and overall functionality in real-world use. Additionally, the proponents tested the website across different browsers and devices to determine its compatibility and ensure consistent user experience. The tables below show the results of different tests conducted by the proponents.

2.5.1 Hardware Compatibility for Desktop and Mobile

Table 13 Hardware Compatibility for Desktop and Mobile

Hardware Specs	Activities	Result
8GB RAM, Intel Core i5 12th Generation (Desktop)	System's Navigation, User registration, chatbot conversation, location detection, meal recommendations, rating and feedback	The website works smoothly with no performance issue.
4GB RAM, Intel Core i5 10th Generation (Desktop)	System's navigation, User registration, chatbot conversation, location detection, meal recommendations, rating and feedback	There is a slight delay when loading the homepage and Menu from the overview but overall, it performs smoothly.
4GB RAM, Android 9 (Mobile)	Accessing the homepage, chatbot interaction, location detection, Overview of food establishment	The website cannot fully load the homepage due to limited system resources.
4GB RAM, Android 10 (Mobile)	System's navigation, user registration, chatbot conversation, meal recommendation, location detection, rating and feedback	Experiences minor delays when loading but runs smoothly after initialization.
8GB RAM, Android 10 (Mobile)	System's navigation, user registration, chatbot conversation, meal recommendation, location detection, rating and feedback	The website performs smoothly with minimal lag and stable functionality.

Table 13 presents the compatibility results of both desktop and mobile devices when performing the activities on the website.

2.5.2 Web browser Compatibility Testing

Table 14 Hardware Compatibility for Desktop and Mobile

Module Testing on different web browser					
Functions /Modules	Web Browser				
	Google Chrome	Mozilla Fox	Microsoft Edge	Brave Browser	Opera Browser
Sign in / Sign up	✓	✓	✓	✓	✓
Chatbot Conversation	✓	✓	✓	✓	✓
Personalized Meal Recommendation	✓	✓	✓	✓	✓
Location Detection (Google Map Integration)	✓	✓	✓	✓	✓
View Nearby Restaurants	✓	✓	✓	✓	✓
Food establishment Directions	✓	✓	✓	✓	✓
Update User Profile (Preferences, Allergies)	✓	✓	✓	✓	✓
View Meal Details	✓	✓	✓	✓	✓
Submit Ratings	✓	✓	✓	✓	✓
View Chat History	✓	✓	✓	✓	✓
Responsive Design (Mobile & Desktop)	✓	✓	✓	✓	✓
User's Favorite Food Establishments	✓	✓	✓	✓	✓
Health Mode and General Mode	✓	✓	✓	✓	✓
Generalized Query	✓	✓	✓	✓	✓
Personalized Query	✓	✓	✓	✓	✓

The table shows the five (5) different browsers where the website was tested using the major activities of the system.

2.5.3 Internet Connectivity Testing

Table 15 Internet Connectivity Testing

Activities	DownStream	UpStream	Response Time
Login	26.05 Mbps	18.90 Mbps	12 ms
User Registration	26.12 Mbps	19.05 Mbps	13 ms
Onboarding	25.14 Mbps	18.09 Mbps	13 ms
Chat	24.97 Mbps	17.82 Mbps	16 ms
Meal Results	24.95 Mbps	17.60 Mbps	16 ms
View Details	24.63 Mbps	17.65 Mbps	16 ms
Meal Menu	28.12 Mbps	17.65 Mbps	17 ms
Directions	23.88 Mbps	17.15 Mbps	17 ms
User Favorites	25.02 Mbps	18.05 Mbps	16 ms
Locate Nearby Restaurant	25.80 Mbps	17.50 Mbps	20 ms
User Profile	25.30 Mbps	18.10 Mbps	16 ms
Change Preferences	25.10 Mbps	18.00 Mbps	14 ms
Rate and Review Meals	25.73 Mbps	18.40 Mbps	16 ms

The table shows the response time when accessing the activities of the website

2.5.4 Web Internet Performance Tesing

The web internet performance testing is the process of rating the responsiveness, speed, and overall performance of the website under real life conditions to ensure a better user experience. This testing helps identify potential issues such as delays or connection errors that could affect system efficiency and user satisfaction.

2.5.4.1 PageSpeed Performance Testing

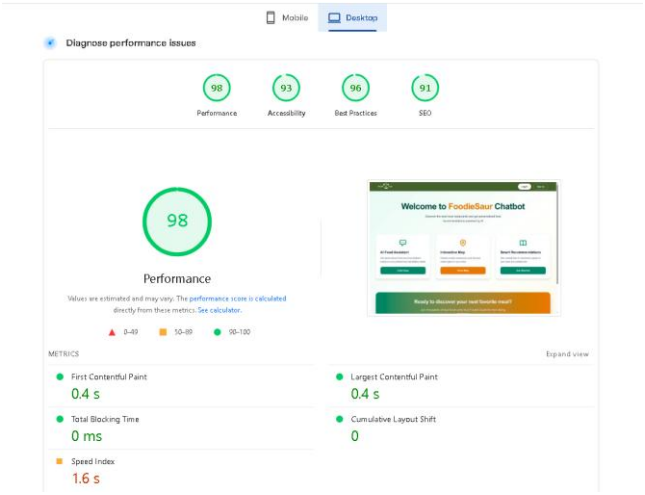


Figure 43 PageSpeed Performance Result

The website PageSpeed Insights shows an impressive performance rate, with metrics indicating a First Contentful Paint of 0.4 seconds and a Total Blocking Time of 0 milliseconds, providing users with a highly responsive experience. Additionally, the Cumulative Layout Shift score of 0 means the layout remains stable without sudden changes, contributing to a positive user impression.

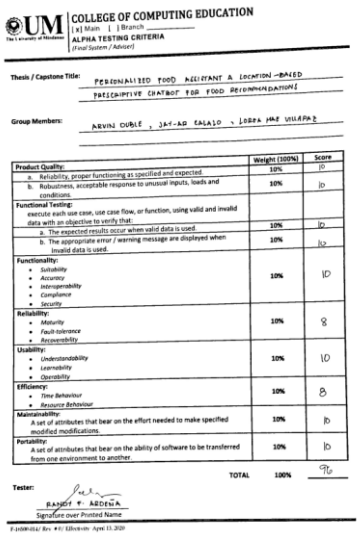
2.5.5 Mobile Screen Resolution Testing

Table 16 Mobile Screen Resolution Testing

Activities	Result	Remarks
Chatbot Interface, Recommendation Results, Restaurant Details, and User's Saved Food establishment, Rate and Review	The panel adjusts properly to various screen sizes, and it can drag up and down and it keeps the button and input fields visible and functional.	PASSED
Map Directions and View	The map is interactive and displays the distance between the user and the designated location.	PASSED
User Profile and Preferences Setup	The setup options diet, health condition, location and budget along with the search bar are functional and can save user changes.	PASSED

This test ensures that the website is mobile-responsive across different screen resolutions. The activity verifies that the buttons, input fields, and other components adjust properly, providing a smooth experience for all users.

2.5.5 Alpha Testing



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[x] Main [ ] Branch _____
ALPHA TESTING CRITERIA
(Final System / Adviser)

Thesis / Capstone Title: PERSONALIZED FOOD ASSISTANT A LOCATION-BASED PRESCRIPTIVE CHATBOT FOR FOOD RECOMMENDATIONS

Group Members: ARVIN DUBLE , JAY-AR CALALO , LOREA MAE VILLAPAZ

	Weight (100%)	Score
Product Quality: a. Reliability, proper functioning as specified and expected.	10%	10
b. Robustness, acceptable response to unusual inputs, loads and conditions.	10%	10
Functional Testing: execute each use case, use case flow, or function, using valid and invalid data with an objective to verify that: a. The expected results occur when valid data is used.	10%	10
b. The appropriate error / warning message are displayed when invalid data is used.	10%	10
Functionality: • Suitability • Accuracy • Interoperability • Compliance • Security	10%	10
Reliability: • Maturity • Fault-tolerance • Recoverability	10%	8
Usability: • Understandability • Learnability • Operability	10%	10
Efficiency: • Time Behaviour • Resource Behaviour	10%	8
Maintainability: A set of attributes that bear on the effort needed to make specified modified modifications.	10%	10
Portability: A set of attributes that bear on the ability of software to be transferred from one environment to another.	10%	10
TOTAL	100%	96

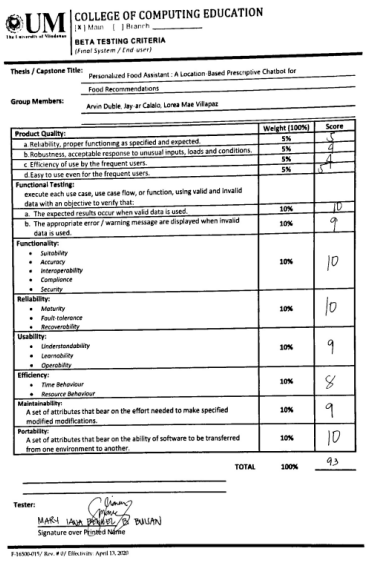
Tester: RANDY F. ARDEÑA
Signature over Printed Name

F-16300-015/ Rev. # 0/ Effectivity: April 13, 2020

Figure 44 Alpha Testing

The figure above shows the alpha testing result of the proposed project. The Website receives a total score of 95% that was done by the proponents and administered by their adviser, Professor Randy F. Ardeña.

2.5.6 Beta Testing



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[x] Main [ ] Branch _____
BETA TESTING CRITERIA
(Final System / End user)

Thesis / Capstone Title: Personalized Food Assistant : A Location Based Prescriptive Chatbot for Food Recommendations

Group Members: Arvin Duble, Jay-ar Calalo, Lorea Mae Villapaz

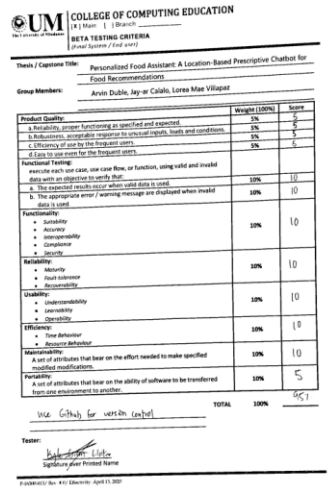
	Weight (100%)	Score
Product Quality: a. Reliability, proper functioning as specified and expected.	5%	5
b. Robustness, acceptable response to unusual inputs, loads and conditions.	5%	5
c. Efficiency of use by the frequent users.	5%	5
d. Easy to use even for the frequent users.	5%	5
Functional Testing: execute each use case, use case flow, or function, using valid and invalid data with an objective to verify that: a. The expected results occur when valid data is used.	10%	10
b. The appropriate error / warning message are displayed when invalid data is used.	10%	9
Functionality: • Suitability • Accuracy • Interoperability • Compliance • Security	10%	10
Reliability: • Maturity • Fault-tolerance • Recoverability	10%	10
Usability: • Understandability • Learnability • Operability	10%	9
Efficiency: • Time Behaviour • Resource Behaviour	10%	8
Maintainability: A set of attributes that bear on the effort needed to make specified modified modifications.	10%	9
Portability: A set of attributes that bear on the ability of software to be transferred from one environment to another.	10%	10
TOTAL	100%	93

Tester: MARY IANA BENNEL B. BUISAN
Signature over Printed Name

F-16300-015/ Rev. # 0/ Effectivity: April 13, 2020

Figure 45 Beta Testing of Prof. Mary Iana Bennel B. Buisan

The figure above shows the results of the proposed project's beta testing, which was participated in by Prof. Mary Iana Bennel B. Buisan. The lowest score was in efficiency due to a weak internet connection, but the other scores met the criteria. Overall, the website received a total score of 93%.



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[x] Main [ ] Branch _____
BETA TESTING CRITERIA
(Final System / End user)

Thesis / Capstone Title: Personalized Food Assistant: A Location-Based Prescriptive Chatbot for Food Recommendations

Group Members: Arvin Duble, Jay-ar Calalo, Lorea Mae Villapaz

	Weight (100%)	Score
Product Quality: a. Reliability, proper functioning as specified and expected.	5%	5
b. Robustness, acceptable response to unusual inputs, loads and conditions.	5%	5
c. Efficiency of use by the frequent users.	5%	5
d. Easy to use even for the frequent users.	5%	5
Functional Testing: execute each use case, use case flow, or function, using valid and invalid data with an objective to verify that: a. The expected results occur when valid data is used.	10%	10
b. The appropriate error / warning message are displayed when invalid data is used.	10%	10
Functionality: • Suitability • Accuracy • Interoperability • Compliance • Security	10%	10
Reliability: • Maturity • Fault-tolerance • Recoverability	10%	10
Usability: • Understandability • Learnability • Operability	10%	10
Efficiency: • Time Behaviour • Resource Behaviour	10%	10
Maintainability: A set of attributes that bear on the effort needed to make specified modified modifications.	10%	10
Portability: A set of attributes that bear on the ability of software to be transferred from one environment to another.	10%	5
TOTAL	100%	95

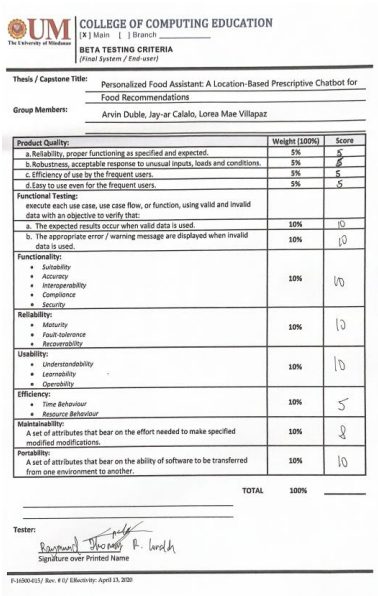
use Github for version control

Tester: Kyle Gian Latina
Signature over Printed Name

F-16300-015/ Rev. # 0/ Effectivity: April 13, 2020

Figure 46 Beta Testing of Prof. Kyle Gian L. Latina

The figure above shows the results of the proposed project's beta testing, which was participated in by Prof. Kyle Gian L. Latina. The lowest score was in portability due to a lack of using GitHub for version control, but the other scores met the criteria. Overall, the website received a total score of 95%.



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BETA TESTING CRITERIA
(Final System / End-user)

Thesis / Capstone Title: Personalized Food Assistant: A Location-Based Prescriptive Chatbot for Food Recommendations

Group Members: Arvin Duble, Jay-ar Calalo, Lorea Mae Villapaz

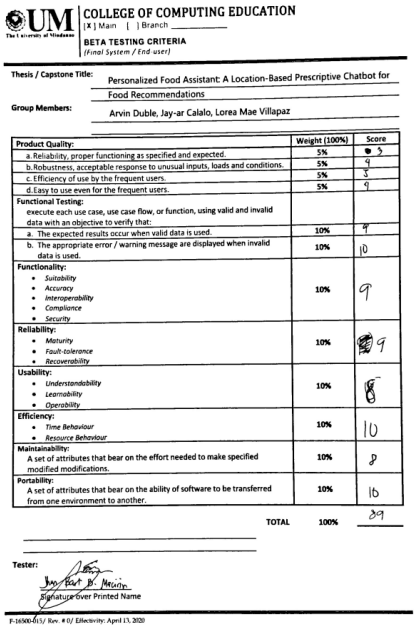
	Weight (100%)	Score
Product Quality: a. Reliability, proper functioning as specified and expected.	5%	5
b. Robustness, acceptable response to unusual inputs, loads and conditions.	5%	5
c. Efficiency of use by the frequent users.	5%	5
d. Easy to use even for the frequent users.	5%	5
Functional Testing: execute each use case, use case flow, or function, using valid and invalid data with an objective to verify that: a. The expected results occur when valid data is used.	10%	10
b. The appropriate error / warning message are displayed when invalid data is used.	10%	10
Functionality: • Suitability • Accuracy • Interoperability • Compliance • Security	10%	10
Reliability: • Maturity • Fault-tolerance • Recoverability	10%	10
Usability: • Understandability • Learnability • Operability	10%	10
Efficiency: • Time Behaviour • Resource Behaviour	10%	5
Maintainability: A set of attributes that bear on the effort needed to make specified modified modifications.	10%	8
Portability: A set of attributes that bear on the ability of software to be transferred from one environment to another.	10%	10
TOTAL	100%	

Tester: Raymund Thomas R. Coralde
Signature over Printed Name

F-16300-015/ Rev. # 0/ Effectivity: April 13, 2020

Figure 47 Beta Testing of Prof. Raymund Thomas R. Coralde

The figure above presents the results of the proposed project's beta testing conducted with Prof. Raymund Thomas R. Coralde. The lowest score was in efficiency, attributed to the need for user interface enhancements. However, the other scores met the required criteria, resulting in an overall website rating of 93%.



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[X] Main [ ] Branch _____
BETA TESTING CRITERIA
(Final System / End-user)

Thesis / Capstone Title: Personalized Food Assistant: A Location-Based Prescriptive Chatbot for Food Recommendations

Group Members: Arvin Duble, Jay-ar Calalo, Lorea Mae Villapaz

Product Quality:	Weight (100%)	Score
a. Reliability, proper functioning as specified and expected.	5%	5
b. Robustness, acceptable response to unusual inputs, loads and conditions.	5%	4
c. Efficiency of use by the frequent users.	5%	5
d. Easy to use even for the frequent users.	5%	4
Functional Testing: execute each use case, use case flow, or function, using valid and invalid data with an objective to verify that:		
a. The expected results occur when valid data is used.	10%	9
b. The appropriate error / warning message are displayed when invalid data is used.	10%	10
Functionality: • *Suitability* • *Accuracy* • *Interoperability* • *Compliance* • *Security*	10%	9
Reliability: • *Maturity* • *Fault-tolerance* • *Recoverability*	10%	9
Usability: • *Understandability* • *Learnability* • *Operability*	10%	8
Efficiency: • *Time Behaviour* • *Resource Behaviour*	10%	10
Maintainability: A set of attributes that bear on the effort needed to make specified modified modifications.	10%	8
Portability: A set of attributes that bear on the ability of software to be transferred from one environment to another.	10%	10
TOTAL	100%	89

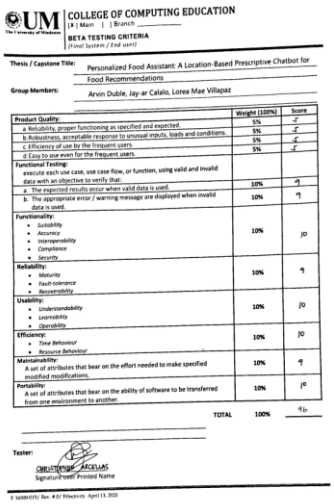
Tester: Jhun Bart B. Macirin
Signature over Printed Name

F-16300-015/ Rev. # 0/ Effectivity: April 13, 2020

Figure 48 Beta Testing of Prof. Jhun Bart B. Macirin

The figure above presents the results of the proposed project's beta testing conducted with Prof. Jhun Bart B. Macirin. The lowest score was in usability and maintainability, due to lack of understanding how the system works as a first-time user. However, the other scores met the required criteria, resulting in an overall website rating of 89%.

2.5.6.1 Beta Testing of Target User



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[X] Main [ ] Branch _____
BETA TESTING CRITERIA
(Final System / End-user)

Thesis / Capstone Title: Personalized Food Assistant: A Location-Based Prescriptive Chatbot for Food Recommendations

Group Members: Arvin Duble, Jay-ar Calalo, Lorea Mae Villapaz

Product Quality:	Weight (100%)	Score
a. Reliability, proper functioning as specified and expected.	5%	5
b. Robustness, acceptable response to unusual inputs, loads and conditions.	5%	5
c. Efficiency of use by the frequent users.	5%	5
d. Easy to use even for the frequent users.	5%	5
Functional Testing: execute each use case, use case flow, or function, using valid and invalid data with an objective to verify that:		
a. The expected results occur when valid data is used.	10%	9
b. The appropriate error / warning message are displayed when invalid data is used.	10%	9
Functionality: • *Suitability* • *Accuracy* • *Interoperability* • *Compliance* • *Security*	10%	10
Reliability: • *Maturity* • *Fault-tolerance* • *Recoverability*	10%	9
Usability: • *Understandability* • *Learnability* • *Operability*	10%	10
Efficiency: • *Time Behaviour* • *Resource Behaviour*	10%	10
Maintainability: A set of attributes that bear on the effort needed to make specified modified modifications.	10%	9
Portability: A set of attributes that bear on the ability of software to be transferred from one environment to another.	10%	10
TOTAL	100%	96

Tester: CHRISTOPHER ARCELLAS
Signature over Printed Name

F-16300-015/ Rev. # 0/ Effectivity: April 13, 2020

Figure 48 Beta Testing of Engr. Christopher Arcellas

The figure above presents the results of the proposed project's beta testing conducted with Engr. Christopher Arcellas. The lowest score was in reliability, due to the system that relies on internet connection. However, the other scores met the required criteria, resulting in an overall website rating of 96%.

2.5.7 Acceptance Testing

UM The University of Mindanao
foodie saur

CERTIFICATE OF ACCEPTANCE
IS HEREBY GIVEN TO

ARVIN L. DUBLE
JAY-AR CALALO
LOREA MAE L. VILLAPAZ

to certify that the Capstone Project entitled
"Personalized Food Assistant: A Location-Based Prescriptive Chatbot for Food Recommendations"
is accepted and to be utilized by the user.

Granted: November 2, 2025
ENGR. CHRISTOPHER ARCELLAS
Target User

Figure 49 Certificate of Acceptance

The testing for acceptance was carried out in a public setting. The end user, Engr. Christopher Arcellas, a time-conscious individual, accepted and approved the functionality and usability of the website.

2.6 Implementation

2.6.1 Front-end Installation

The front-end implementation was developed using modern web technologies to create a responsive and interactive user interface. The application utilizes Tailwind CSS 4.1.7 as the primary styling framework, providing utility-first CSS classes that enable rapid development and consistent design. The installation process began by setting up Node.js and npm to manage front-end dependencies, followed by creating a package.json file that specifies Tailwind CSS and its CLI tools.

The Tailwind CSS configuration file (tailwind.config.js) was set up to scan all Razor view files (.cshtml) throughout the application, ensuring that any Tailwind classes used in templates are included in the final compiled CSS. The build process was integrated directly into the ASP.NET Core project file, where a build target automatically compiles and minifies the CSS before each application compilation, ensuring developers always work with the latest styles without manual intervention.

Google Maps JavaScript API integration was implemented to provide the full-screen interactive map that serves as the application's primary visual element. The API key was securely stored in the application configuration file, and the necessary scripts were included in the main layout to enable map functionality across all pages. Additionally, the Inter font family from Google Fonts was imported to provide clean, modern typography that enhances readability and maintains a professional appearance.

2.6.2 Network and Domain Configuration

The network configuration involves multiple API integrations that enable core functionality. Google Maps API is configured with an API key stored in the application settings, enabling interactive maps, restaurant searches, and directions. Google Dialogflow API is integrated with project credentials stored securely in the secrets directory, powering the AI conversational interface that understands natural language queries. ScrapingBee API provides additional web scraping capabilities to supplement restaurant information beyond what Google Places API offers.

Authentication supports multiple providers for user convenience. Google OAuth and Facebook OAuth are configured with respective Client IDs and Secrets stored in the configuration file, allowing users to sign in using social accounts. These are supplemented with cookie-based authentication using a 30-minute sliding expiration that extends sessions during active use.

The database connection is established through a SQL Server connection string configured for an Azure SQL Database. The connection includes encryption settings for secure data transmission, trust server certificate configurations, and Multiple Active Result Sets enabled for improved performance with concurrent operations.

Security measures include HTTPS redirection for production, secure cookie settings with HttpOnly flags, and session management with automatic logout after inactivity. The application runs on localhost during development, with publishing profiles configured for web deployment to production servers. The allowed hosts setting is configured for flexible deployment, though it should be restricted to specific domains in production for enhanced security.

2.6.3 Database Setup

The database uses SQL Server on Azure SQL Database, with Entity Framework Core 9.0.5 as the ORM framework using a code-first approach. The schema is managed through 26 migration files that track complete schema evolution, enabling version control and rollback capabilities.

The database schema is organized into logical groups. Core user tables include Users for account information with OAuth support and BCrypt password hashing, UserProfiles for extended user details including dietary preferences and health conditions, ChatSessions for tracking conversations, and ChatMessages for storing message history.

Food and preference tables form the recommendation foundation. The FoodTypes master table contains over 1,154 food types classified by meal type. The UserFoodTypes table links users to preferences with 1-10 preference scores. DietaryRestrictions and UserDietaryRestrictions tables manage dietary needs, while HealthConditions and UserHealthConditions tables handle health-related constraints for recommendations.

Behavioral learning tables enable the prescriptive analytics engine. The UserBehaviors table tracks comprehensive interactions including searches, selections, and feedback. The UserPreferencePatterns table stores learned patterns with confidence scores for cuisine preferences, time patterns, and price ranges. The UserFavoriteRestaurants table manages saved restaurants with coordinates, ratings, and timestamps.

The application automatically initializes the database on startup. Migrations are applied automatically to ensure schema currency. The system seeds initial data including over 40 food types across multiple cuisines, 15 common dietary restrictions, 15 health conditions, and sample preference patterns for testing. Database configuration includes decimal precision settings for coordinates and confidence scores, foreign key relationships with cascade restrictions, and indexed relationships for optimal query performance.

2.6.4 User's Manual and Turnover activities

The user manual guides users through application features. Registration supports email/password or social login through Google/Facebook. The login system uses secure cookie-based sessions, and the application request's location permission for personalized recommendations.

The main interface consists of three components working together. The full-screen interactive map displays nearby restaurants as markers. The left search panel features Google Places autocomplete, displaying results as cards with photo carousels, ratings, cuisine types, and distance indicators. Action buttons enable saving favorites, sharing via native Web Share API, and opening Google Maps directions. The right chat panel houses the FoodieSaur AI assistant that processes natural language queries and provides prescriptive recommendations based on user preferences, health conditions, and behavioral patterns. The favorites panel displays saved restaurants with quick access to navigation and sharing.

The preferences page allows users to customize recommendations by rating food types on a 1-10 scale, selecting dietary restrictions, specifying health conditions, and indicating meal type preferences. The chat interface supports various query types including simple requests like "I want pizza for dinner," health-focused queries like "Show me healthy options," constraint-based searches like "What restaurants match my dietary restrictions," and geographic queries like "Find vegetarian restaurants within 2km.

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